



BIOLOGÍA DEL COMPORTAMIENTO

Introducción a la investigación en Neurociencias

turno vespertino 2026

Mariana Imperatori
Pablo Koss

[Holmes:] No tengo los datos todavía. Es un error garrafal teorizar antes de tener los datos. Sin darse cuenta, uno empieza a retorcer los hechos para que concuerden con las teorías, en lugar de que las teorías se ajusten a los hechos.

A. CONAN DOYLE (1891-1989)

La perfección es el gran enemigo
de un buen principio.

ANÓNIMO

Cuanto más dilatada es la isla del conocimiento,
más extensa es la costa de lo
desconocido.

JOHN DONNE

Ciencia es la voluntad de aceptar los
hechos aunque se opongan a los deseos.
B. F. SKINNER (1953)

Kandel: La ciencia del ¿cerebro?

toda conducta es el resultado de la función cerebral. Lo que conocemos comúnmente como mente es un conjunto de operaciones que el cerebro lleva a cabo.



Kandel ER, Schwartz JH, Jessell TM, Siegelbaum SA, Hudspeth AJ, editores. Principios de neurociencia. 5ª ed. Madrid: McGraw-Hill Interamericana de España; 2013.

Neurociencias

Abordaje multidisciplinario de las bases biológicas del comportamiento

Disciplinas:

- *Neuroanatomía
- *Neuroquímica
- *Neurofisiología
- *Neuropsicología

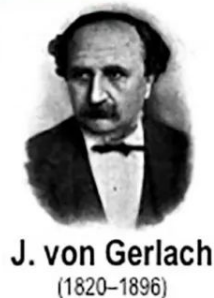
Ramas:

- *Neurociencia Molecular
- *Neurociencia Cognitiva
- *Neurociencia Clínica
- *Neurociencia Computacional
- *Neurociencia del Desarrollo
- *Neurociencia Cultural

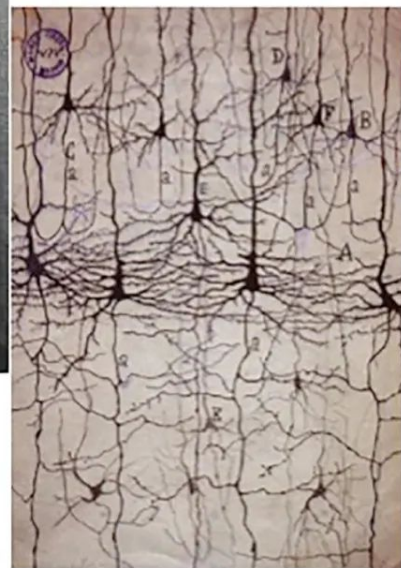


Bueno vamo a investiga'

A Reticularists



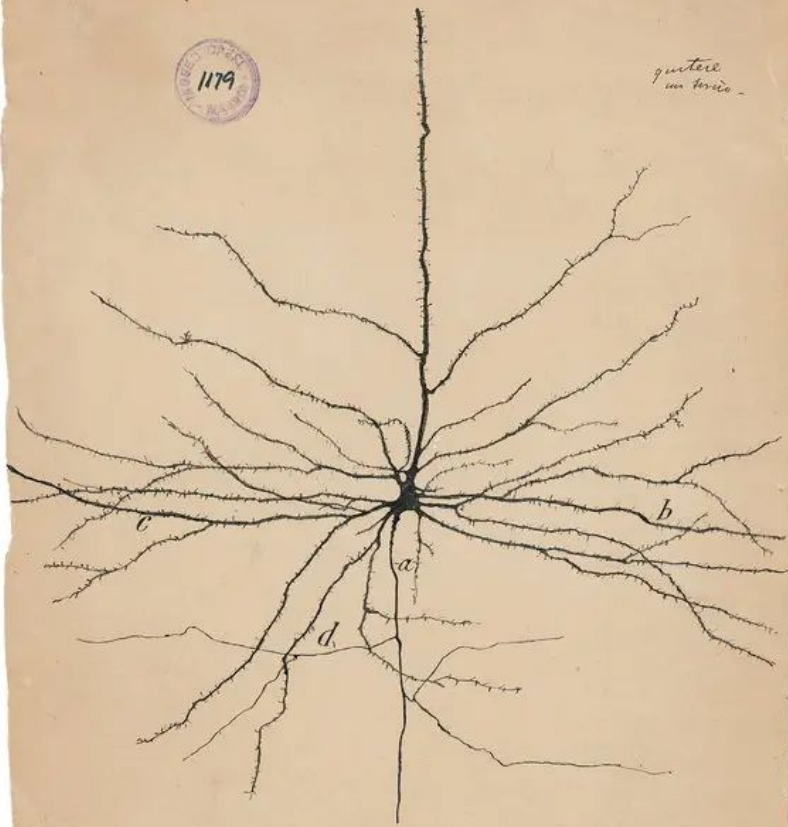
B Neuronists



Nobel Prize
in Physiology or Medicine
1906



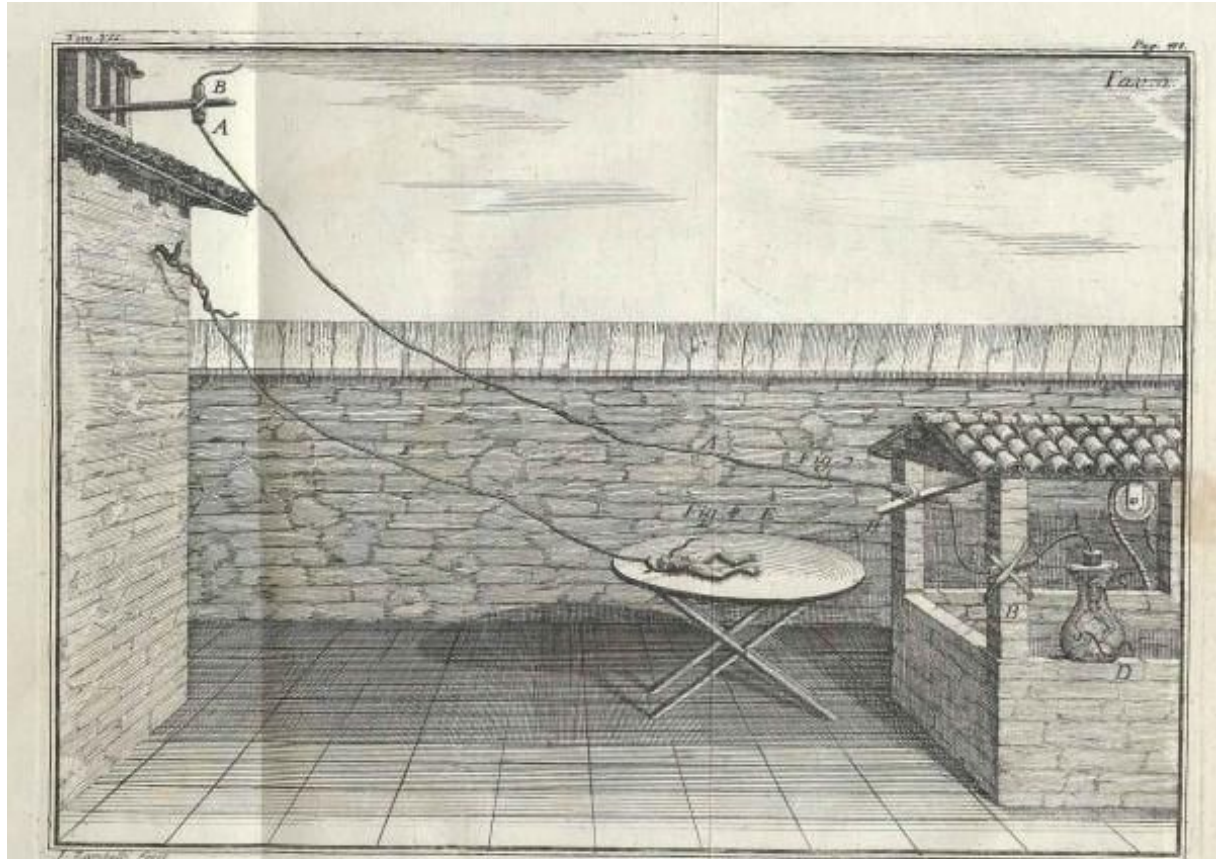
*gaster
un solo*





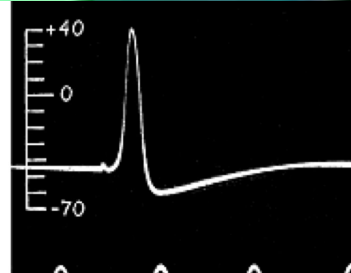
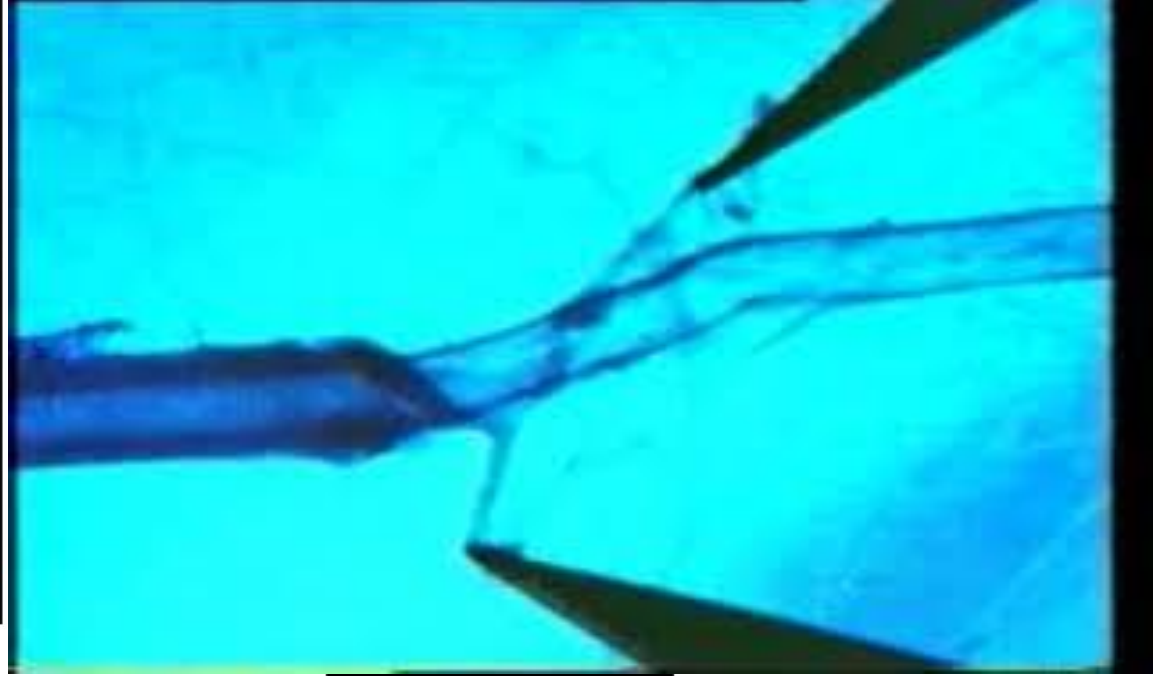
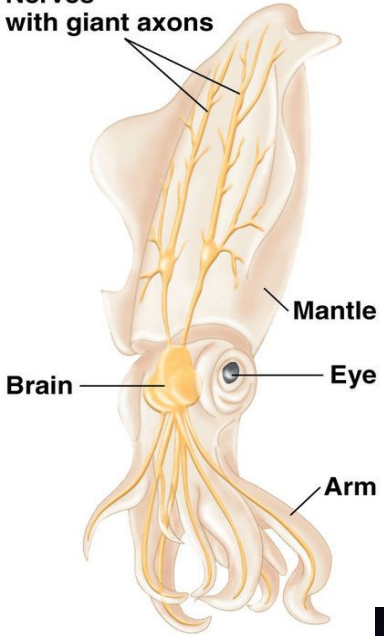
HD

Luigi Galvani (1771): “electricidad animal”

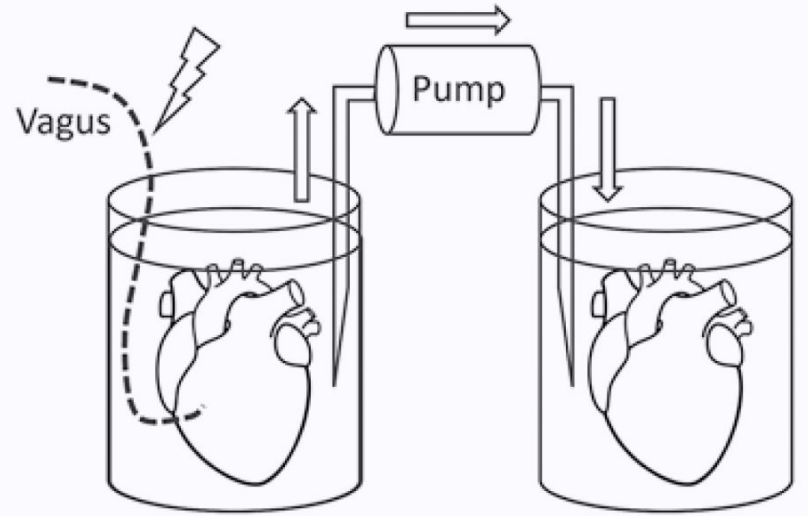


Hodgkin y Huxley (1963): potencial de acción

Nerves
with giant axons



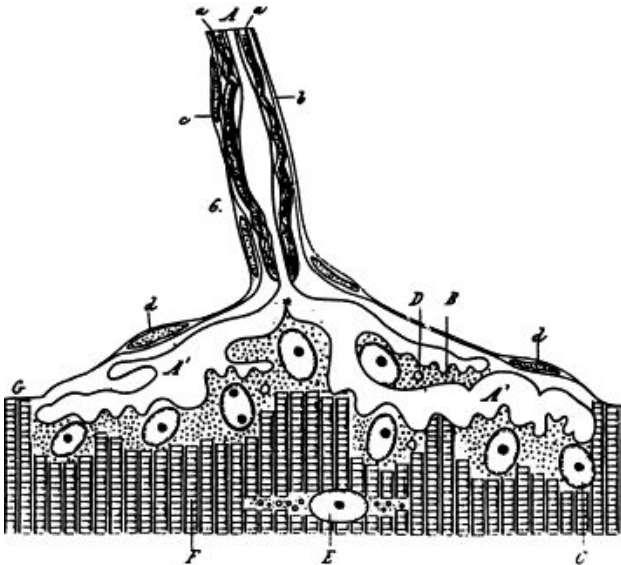
Otto Loewi (1926): Vagusstoff - Comunicación no eléctrica



Without stimulation	Heart rate		
With stimulation	Heart rate		

"En la noche del Sábado de Pascua de 1921, me desperté, encendí la luz y anoté algunas ideas en un diminuto trozo de papel. Luego volví a quedarme dormido. A las seis de la mañana recordé que durante la noche había escrito algo muy importante, pero fui incapaz de descifrar el garabato. Ese domingo fue el día más desesperante de toda mi vida científica. Durante la noche siguiente, sin embargo, volví a despertarme, a las tres de la madrugada, y recordé lo que era. Esta vez no corrí ningún riesgo; me levanté inmediatamente, fui al laboratorio, realicé el experimento con el corazón de rana descrito anteriormente, y a las cinco de la madrugada la transmisión química del impulso nervioso quedó concluyentemente demostrada." Otto Loewi

Charles Sherrington (1906): Sinapsis



"Hasta donde llega nuestro conocimiento actual, nos vemos llevados a pensar que la punta de una ramita de la arborización no es continua, sino que está meramente en contacto con la sustancia de la dendrita o el cuerpo celular sobre el que incide. Tal conexión especial de una célula nerviosa con otra podría denominarse sinapsis."

Fatt y Katz (1952): Neurotransmisión química

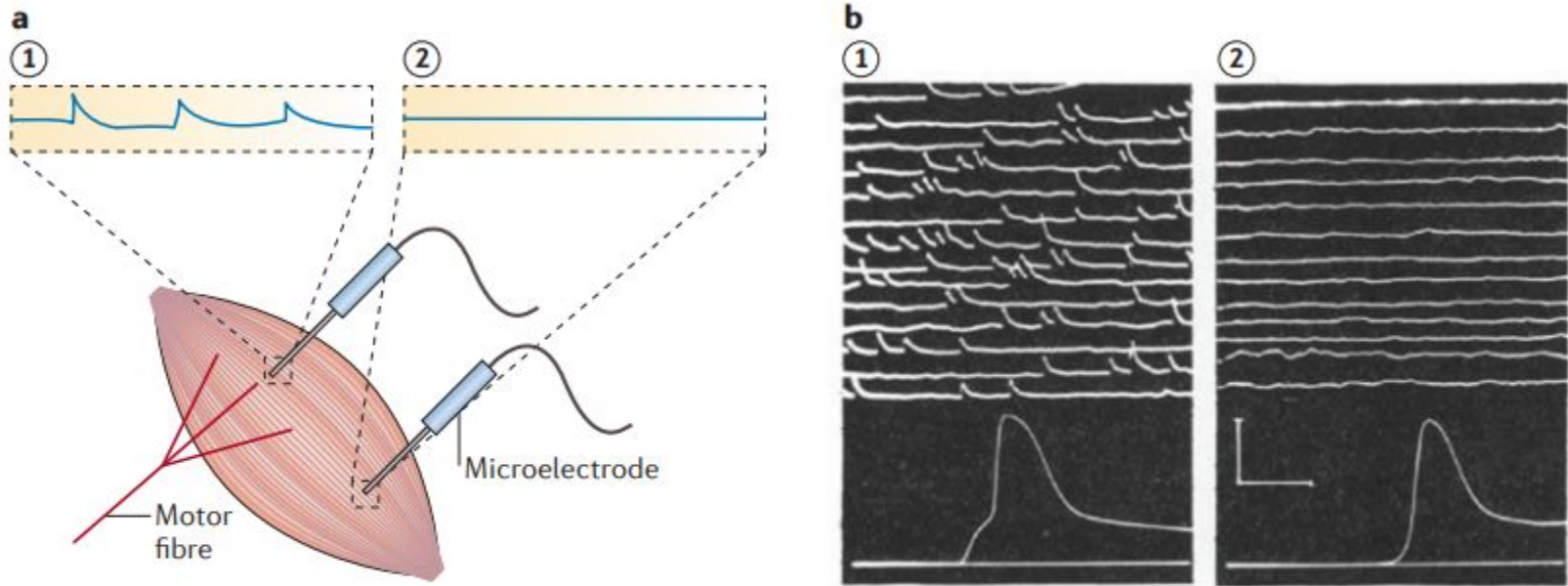
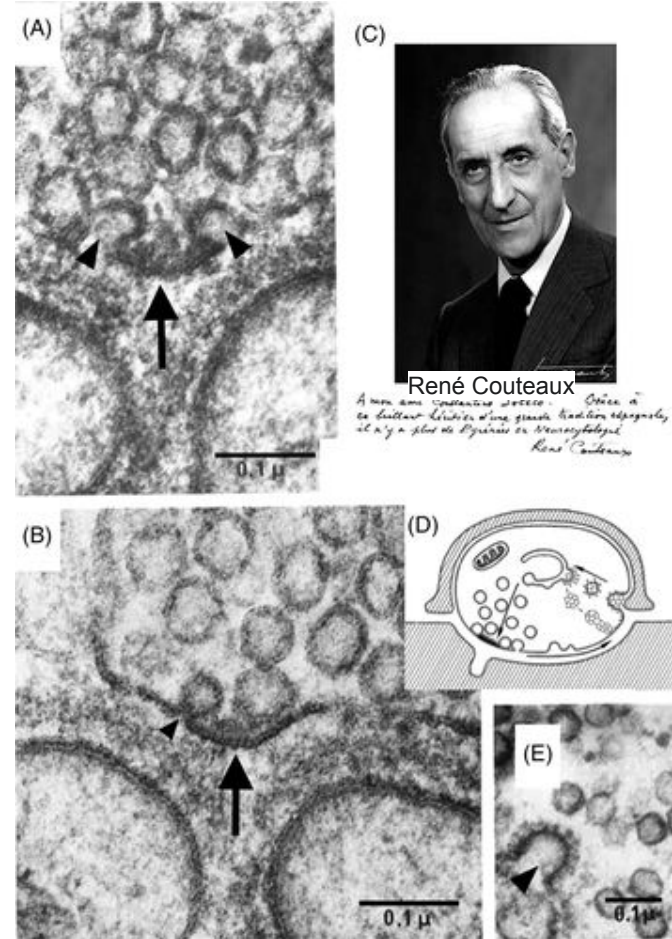
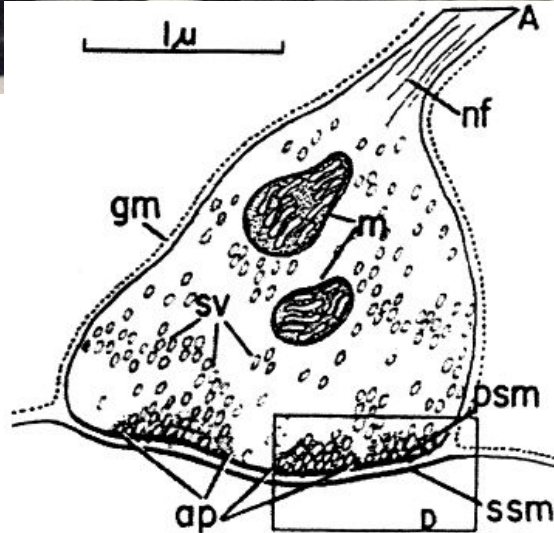
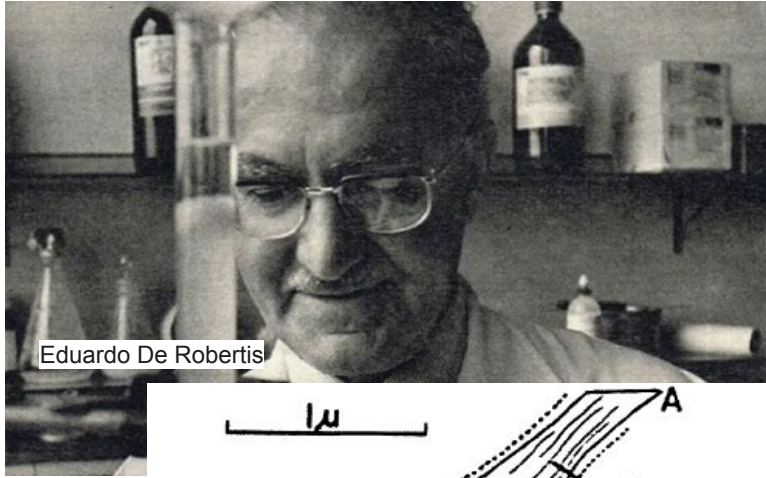
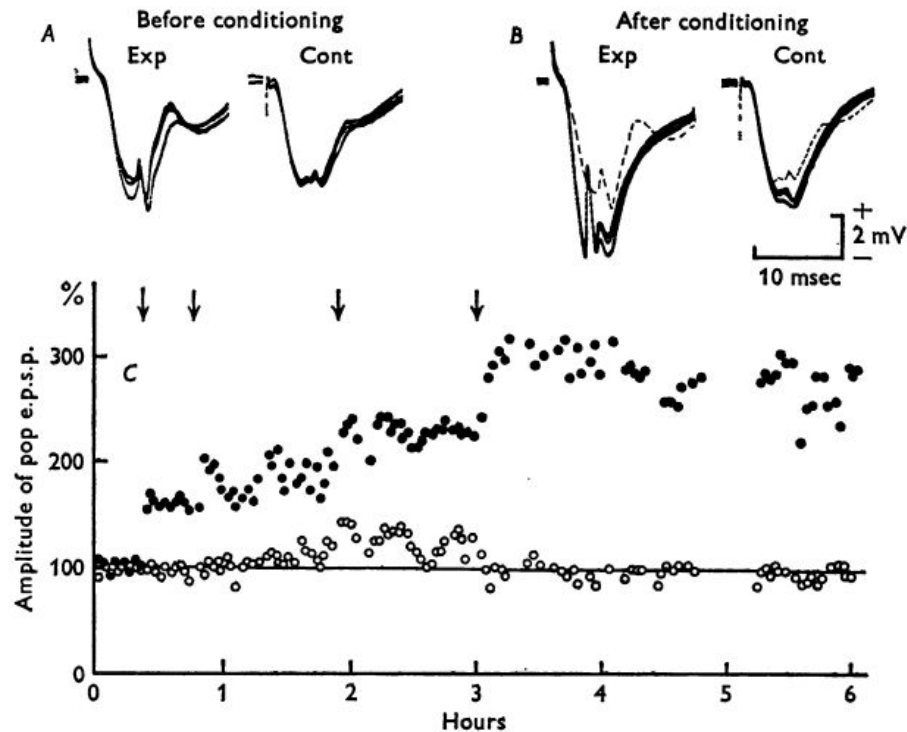
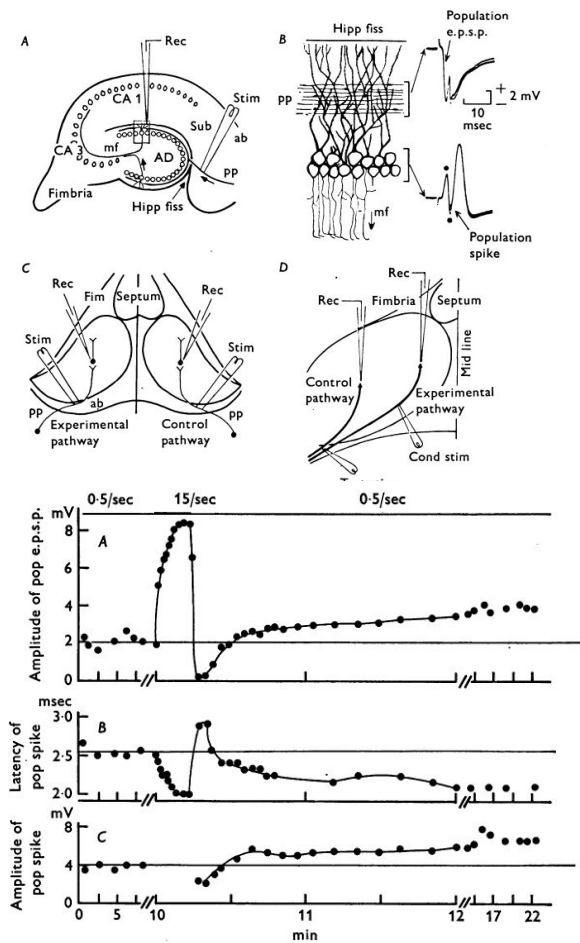


Figure 1 | **The earliest recordings of spontaneous synaptic activity.** **a** | The neuromuscular junction preparation used by Bernard Katz and colleagues, in which they initially described spontaneous cholinergic neurotransmitter release events, is shown. **b** | Intracellular recordings near the sites of nerve innervation (1), but not recordings at distal regions (2), showed random spontaneous voltage fluctuations that were sensitive to cholinergic agents, nerve denervation and osmotic pressure². Part **b** reproduced from Fatt, P. & Katz, B. Spontaneous subthreshold activity at motor nerve endings. *J. Physiol.* 117, 109–128 (1952). John Wiley & Sons.

De Robertis y Couteaux (1950): Vesículas sinápticas



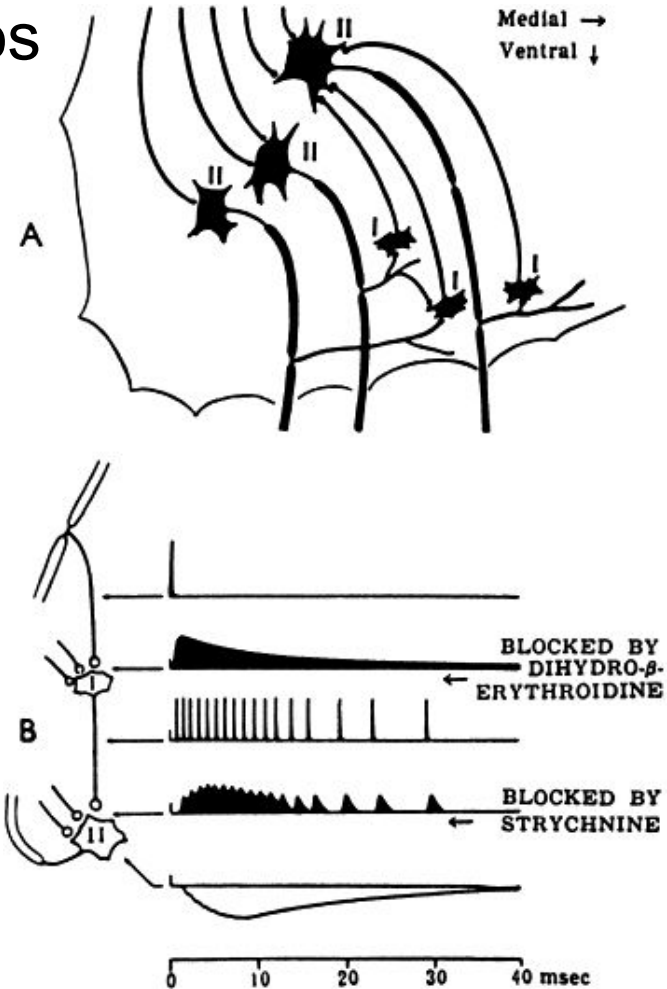
Bliss y Lomo (1973): Plasticidad eléctrica - LTP



Eccles (1959): Microcircuitos - Inhibitorios

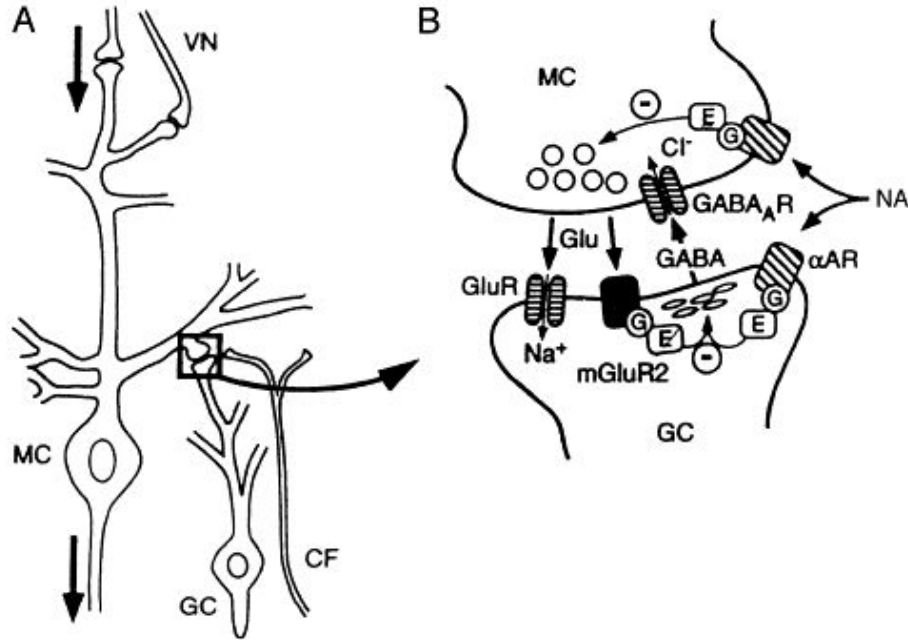
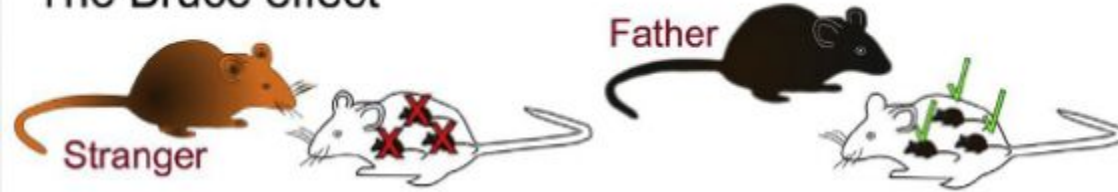
(A) Motoneuronas, II, e interneuronas inhibitoras de Renshaw, I.

(B) Análisis intracelular que muestra un potencial de acción inicial en una colateral del axón de una motoneurona, un EPSP en una célula de Renshaw I; bloqueado por dihidro- β -eritroidina (antagonista Ach) que conduce a una descarga intensa de potencial de acción, una secuencia rápida de breves despolarizaciones de las terminales axónicas de la célula de Renshaw que liberan un transmisor inhibitorio para actuar sobre la motoneurona II; bloqueado por estricnina (antagonista Glicina), y un potencial postsináptico inhibitorio hiperpolarizante de larga duración (IPSP) en la motoneurona.



Nakanishi (1994): Circuito asociado a comportamiento

The Bruce effect



(A) Mitral–granule cell relations in the accessory olfactory bulb: VN, vomeronasal nerve; MC, mitral cell; GC, granule cell; CF, centrifugal fibre from Locus Coeruleus.

(B) Molecular components of the dendrodendritic synapses that underlie sensory processing and olfactory memory. E, effector; G, G protein; GABA, gamma-aminobutyric acid; GABA_AR, GABA_A receptor; Glu, ionotropic glutamate receptor; mGluR2, metabotropic glutamate receptor; αAR, alpha adrenoreceptor; NA, noradrenaline.

Nakanishi (1994): Circuito asociado a comportamiento

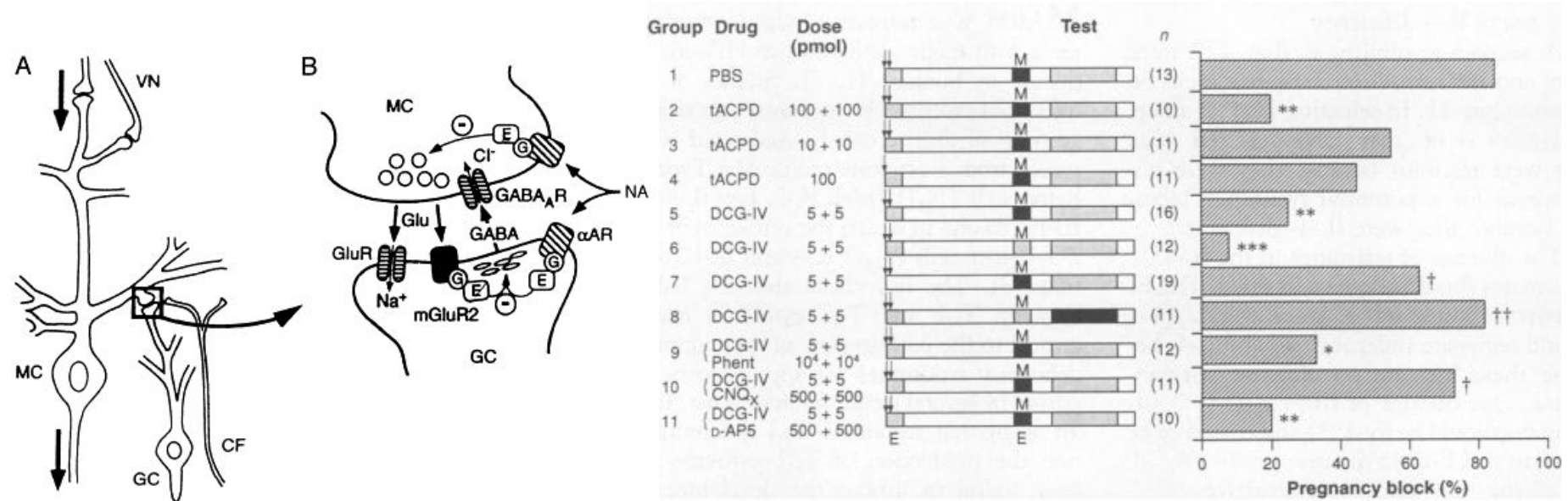
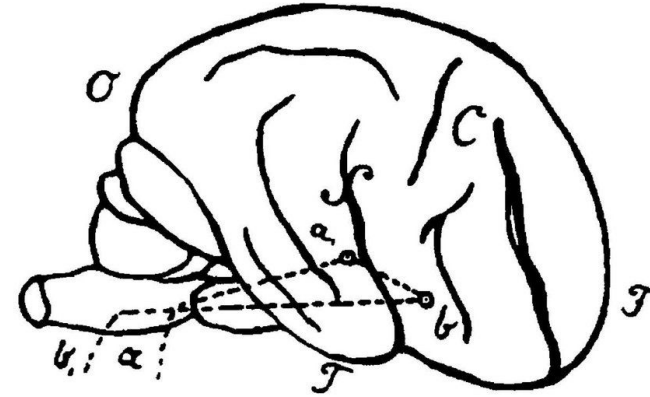
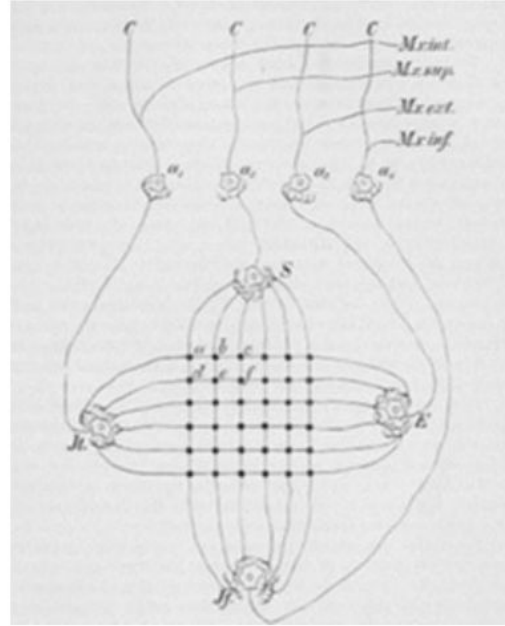
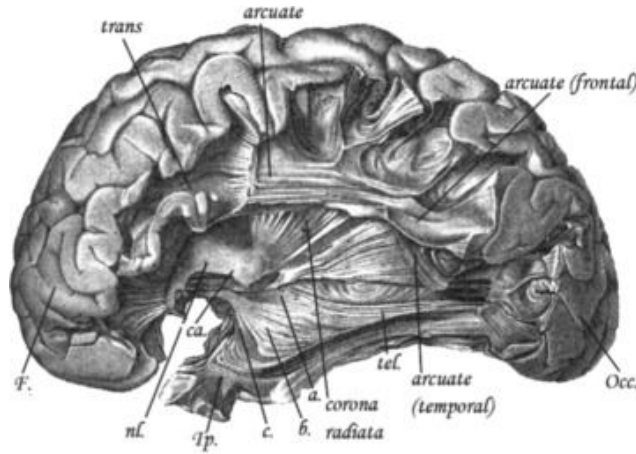


Fig. 2. Olfactory memory formation by mGluR activation in the AOB. Failure of the test exposure to male pheromones to block pregnancy was used to evaluate the formation of an olfactory recognition memory (9). M, mating; arrows, drug or phosphate-buffered saline infusions; shaded blocks, exposure to pheromones from the BALB/c male; filled blocks, exposure to pheromones from the CBA male; Phent, phentolamine; and E, estrus. The numbers in parentheses indicate numbers of animals tested. * $P < 0.05$, ** $P < 0.01$, and *** $P < 0.005$, when compared with group 1. † $P < 0.05$, †† $P < 0.01$, when compared with group 5. Data were analyzed with the Fisher exact probability test.

Meynert y Wernicke (1860): Macrocircuitos



The association-bundles may be compared to a connecting thread, which enables one image to lift the other, as it were, over the threshold of consciousness. T. Meynert 1865



Disposición 9236 / 2023

ADM.NAC.DE MEDICAMENTOS, ALIMENTOS Y TEC. MEDICA

ADMINISTRACION NACIONAL DE MEDICAMENTOS, ALIMENTOS Y TECNOLOGIA MEDICA

REGIMEN DE BUENAS PRACTICAS EN BIOTERIOS - APROBACION

Sanción: 01-11-2023

Publicada en el Boletín Oficial: [03-11-2023](#)

Número: [35291](#)

Página: 98

1.1. Sólo podrán utilizarse animales para fines de investigación, desarrollo o control biológico luego de haber aplicado con rigurosidad el Principio de las Tres Erres de Russell y Burch.

- Reemplazo de los animales por métodos alternativos que produzcan los mismos resultados, tales como experimentos in vitro, cultivos celulares, modelos inanimados, modelos por computadora, etc.
- Reducción del número de animales utilizando el mínimo número necesario para obtener resultados significativos, y
- Refinamiento de las técnicas, la metodología, el instrumental y el diseño del experimento en sí, para minimizar el malestar y el dolor.

No podrán utilizarse animales para ninguno de los fines mencionados cuando hubiera un método alternativo satisfactorio desde el punto de vista técnico y científico que no requiera del uso de los mismos.

1.2. Cuando sea inevitable el uso de animales vivos, todas las maniobras experimentales deberán ser realizadas evitando o minimizando el sufrimiento y manteniéndolos en condiciones de alojamiento confortables.

Replacement

- Why is it not possible to achieve the objectives of your project without using animals?
- What alternatives have you considered and why are they not suitable? What alternatives will be used in achieving your objectives?

Reduction

- What measures have been or will be taken to ensure that the minimum number of animals will be used in this project?
- Explain the principles of experimental design you will use and any sources of advice you will consult e.g. on statistics.

Refinement

- Explain your choice of species, model(s) and method(s). Explain why they are the most refined for the intended purpose.
- How will you minimise animal suffering in order to achieve your objectives?
- Provide specific justification for any protocols categorised as 'severe'.

13. COMITÉ INSTITUCIONAL DE CUIDADO Y USO DE ANIMALES DE LABORATORIO (CICUAL)

Se recomienda que los establecimientos cuenten con un Comité Institucional para el Cuidado y Uso de los Animales de Laboratorio (CICUAL), que por la experiencia y pericia de sus miembros sea idóneo para promover un trato humanitario a los animales de experimentación y fomentar su reemplazo por otros métodos de investigación, la reducción en el uso de animales y el refinamiento de los procedimientos utilizados. El mismo tiene como finalidad principal la protección y bienestar de los animales utilizados en controles biológicos, investigación y/o experimentación, capacitación y docencia, así como sus condiciones de cría, alojamiento, sanidad y nutrición que se desarrollen dentro del marco de las normas nacionales e internacionales que regulan el uso ético de los mismos.

IN VIVO MODELS

Fruit Fly
(*Drosophila melanogaster*)



Advantage

- Easy to work with
- Short generation time
- Low cost of maintenance

Limitations

- ❖ Genetically distant from human
- ❖ Relatively simple anatomy
- ❖ It has no adaptive immune system

Zebrafish
(*Danio Rerio*)



Advantage

- High reproductive rate
- Development is external
- Genetic similarity to human

Limitations

- ❖ Moderate flexibility
- ❖ Moderate predictivity
- ❖ Moderate translational value

Mouse
(*Mus musculus*)



Advantage

- Complex behavior
- Organ homologous to humans
- Genetic similarity to human

Limitations

- ❖ Very expensive husbandry costs
- ❖ Experimental cycle long
- ❖ Ethical constraints

IN VITRO MODELS



Advantage

- Easy to work with
- Low cost of maintenance
- Short experimental cycle

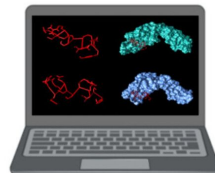
Limitations

- ❖ Simplified system
- ❖ Highly controlled
- ❖ Poor correlation with in-vivo mechanisms

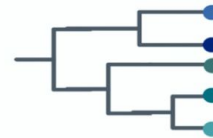
Alternative approaches to animal models



In vitro (cell culture, microscopy) techniques



In silico (computational) modeling



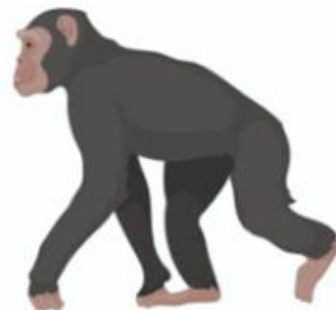
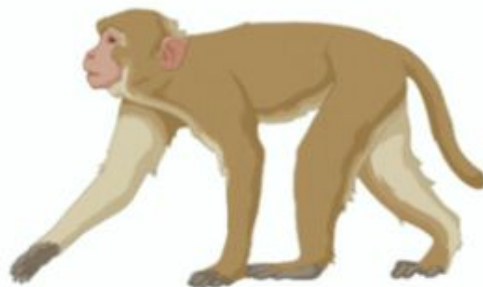
Evolutionary perspective



Biomedical perspective



Ethical perspective



In vitro models

"brain-in-a-dish"



2D neuronal
(co-)cultures



3D organoids



Microfluidic
brain-on-a-chip

In vivo vertebrate models

non-mammalian



Zebrafish



Rodents



Xenografted
rodents



Non-human
primates

Ethically justifiable

Physiological relevance
Complexity
Face validity

NAM use



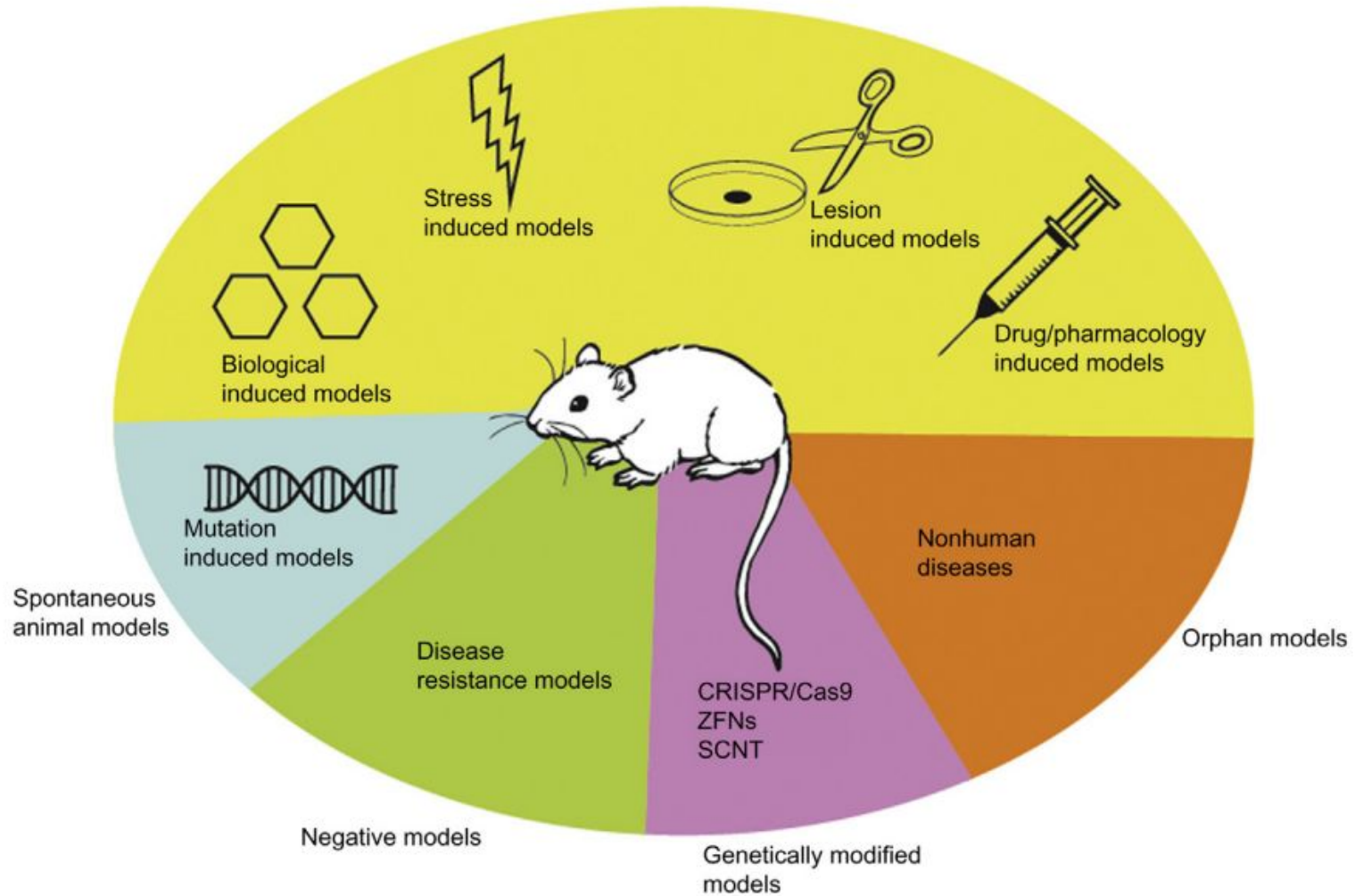
Throughput



Cost-effective
Time-efficient



Induced animal models

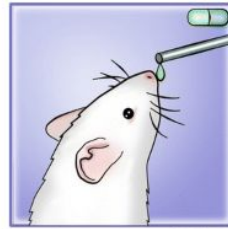




LPS injection



Bulbectomy



Corticosterone drinking



Genetic models



Optogenetic manip.



Maternal separation



Poor mothering

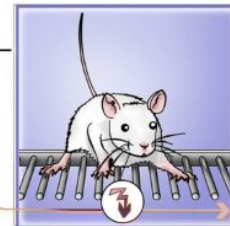
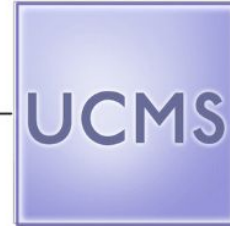
• Early life adversity

• Biological causation

• Adult stress



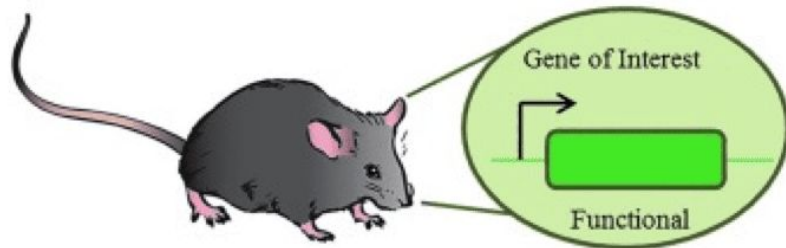
Social defeat



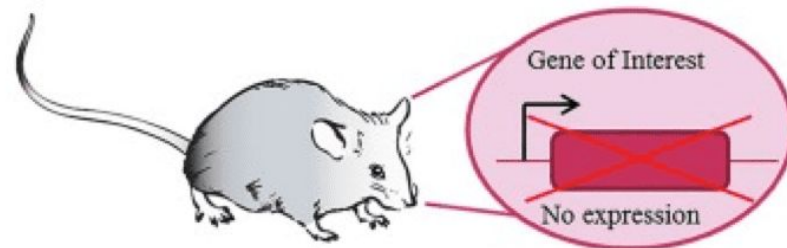
Learned helplessness

ANIMAL MODELS

A Wild type mouse

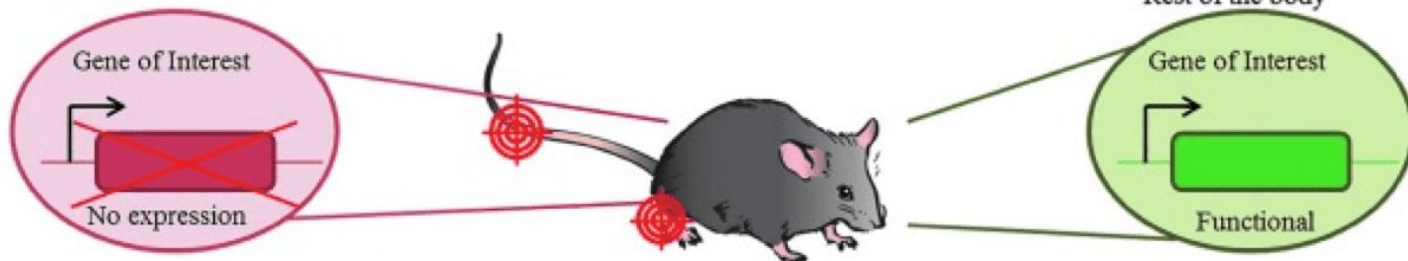


Constitutive Knockout mouse

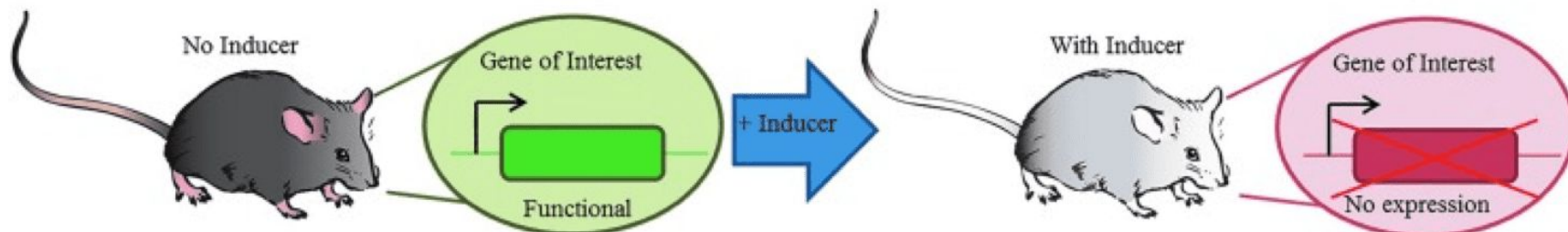


B Tissue-specific Knockout mouse

Targeted tissue (tendons)



C Inducible Knockout mouse



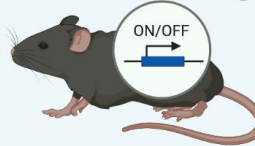
Types of Genetic Mouse Models

Constitutive Knockout Mouse



Constitutive knockout mice are often called "global knockout mice" because *a specific gene has been deleted from every cell* in the entire mouse from the embryo stage.

Conditional Knockout Mouse



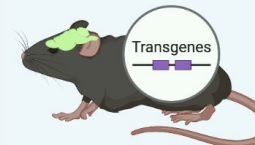
Conditional knockout mice will have the target gene deleted only in certain cell types and/or only after a specific time point in their development. This model allows us more control to decide *when* and *where* a gene is deleted.

Knockin Mouse

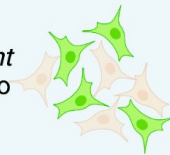


In some cases, mice do not express a certain gene that we would like to study, or the mouse version of a gene is significantly different from the human version. In these cases, we can *add a gene* and create a knockin mouse model.

Reporter Mouse

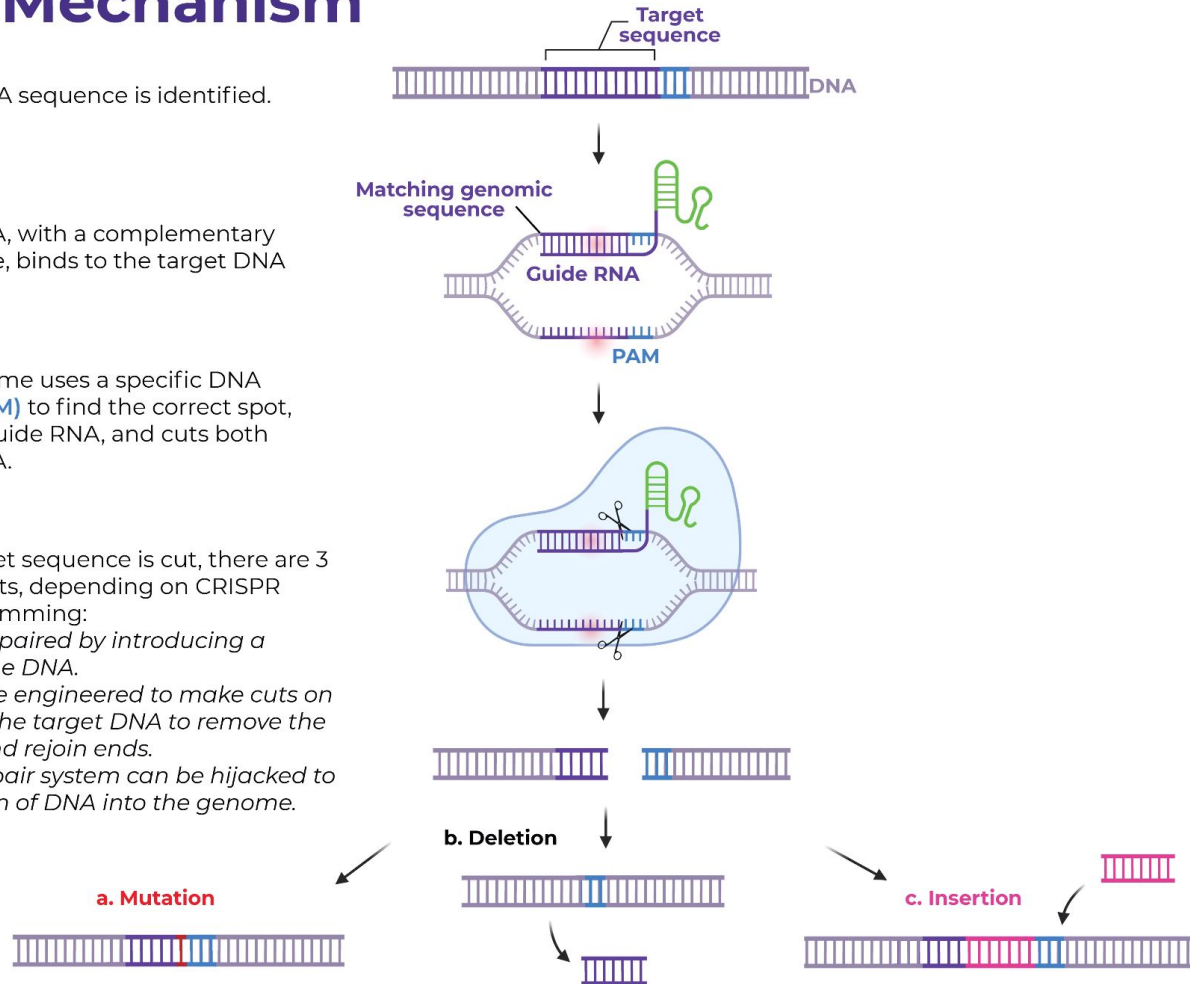


We can *insert genes that produce fluorescent proteins* in a certain subpopulation of cells to help easily identify them, creating what is known as a reporter mouse.

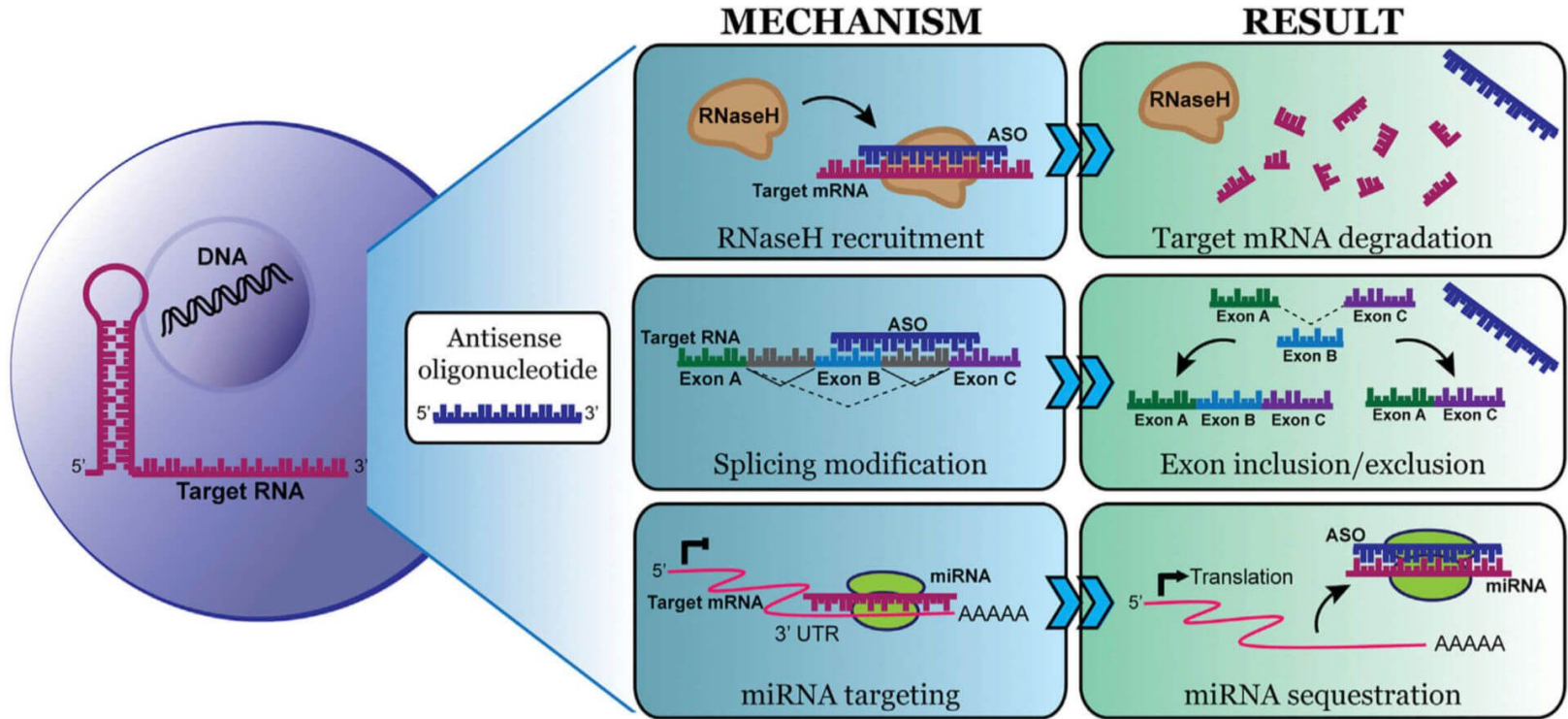


CRISPR Mechanism

- ① The target DNA sequence is identified.
- ② The guide RNA, with a complementary DNA sequence, binds to the target DNA sequence.
- ③ The Cas9 enzyme uses a specific DNA sequence (**PAM**) to find the correct spot, binds to the guide RNA, and cuts both strands of DNA.
- ④ Once the target sequence is cut, there are 3 potential results, depending on CRISPR system programming:
 - a. The cut is repaired by introducing a mutation in the DNA.
 - b. Enzymes are engineered to make cuts on either side of the target DNA to remove the target DNA and rejoin ends.
 - c. The DNA repair system can be hijacked to insert a section of DNA into the genome.



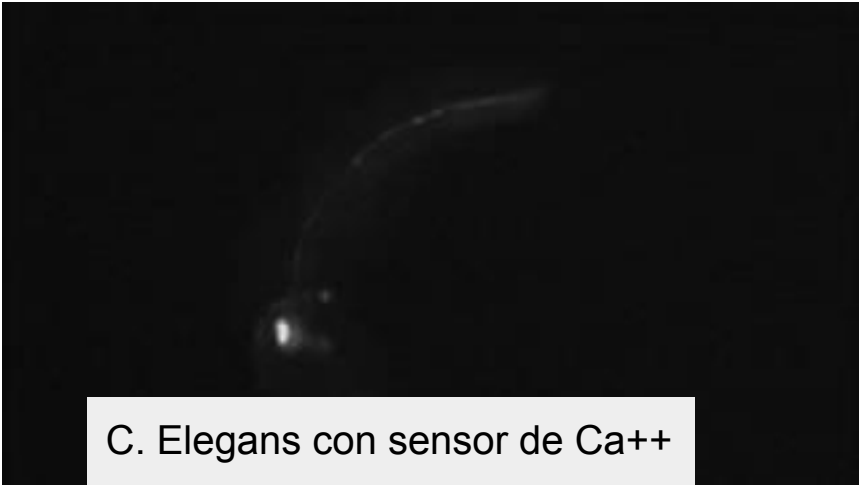
Antisense



¿Cómo modelamos?

- a) Enfermedad de Alzheimer
- b) Ansiedad
- c) Autismo

¿Cómo evaluamos?



A dark, low-contrast image showing a small, bright, circular spot, likely representing a C. elegans worm with a Ca++ sensor.

C. Elegans con sensor de Ca⁺⁺



A photograph of a Y-shaped maze with three white arms extending from a central point. A small, dark object is visible in the top arm.

Y maze: working memory



A photograph of an Elevated Plus Maze (EPM) apparatus. It consists of a horizontal bar with a central platform and two vertical arms. A mouse is visible on the platform. An orange arrow points to the platform. Text on the right side of the image reads "Closed Arm 00:02" and "Mouse took enter in Arm".

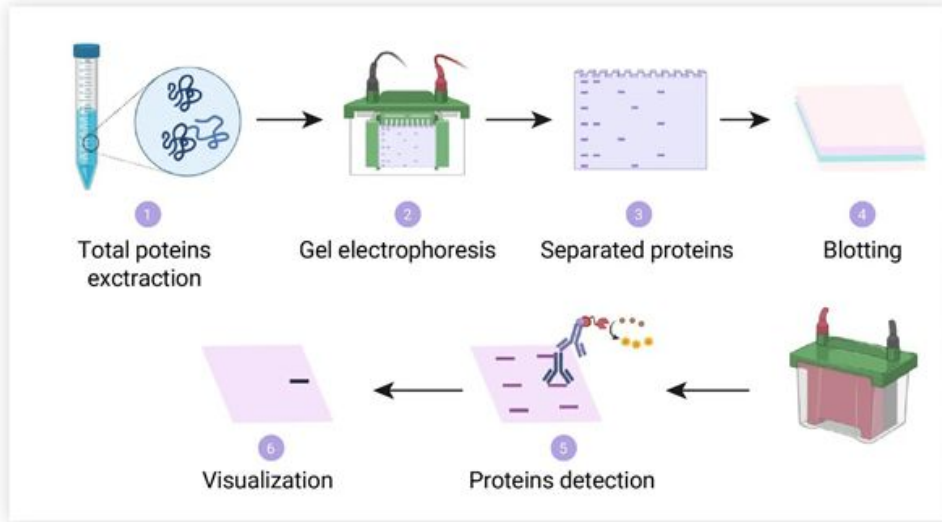
EPM: conductas tipo ansiedad



A photograph of a Morris Water Maze (MWM) apparatus. It shows a circular pool of water with a dark, circular platform in the center. A mouse is visible on the platform.

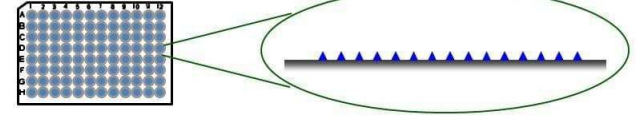
MWM: Aprendizaje y memoria espacial

¿Cómo evaluamos?



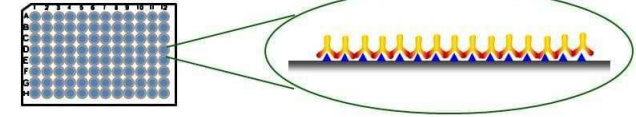
Western Blot

1. Coat antigen onto microplate

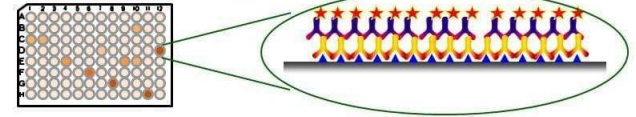


2. Allow protein adsorption and block unoccupied sites with neutral protein

3. Add antibody solution into each well



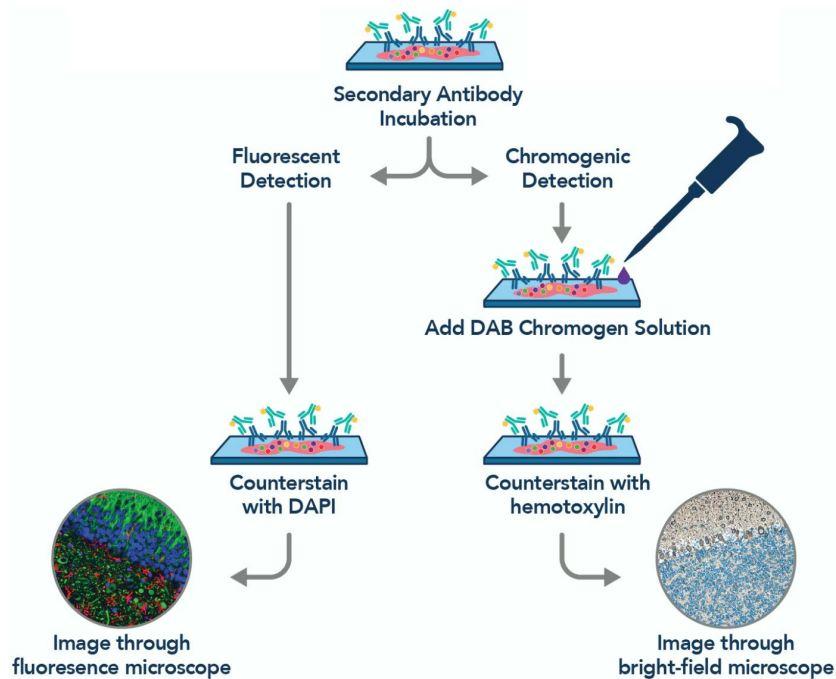
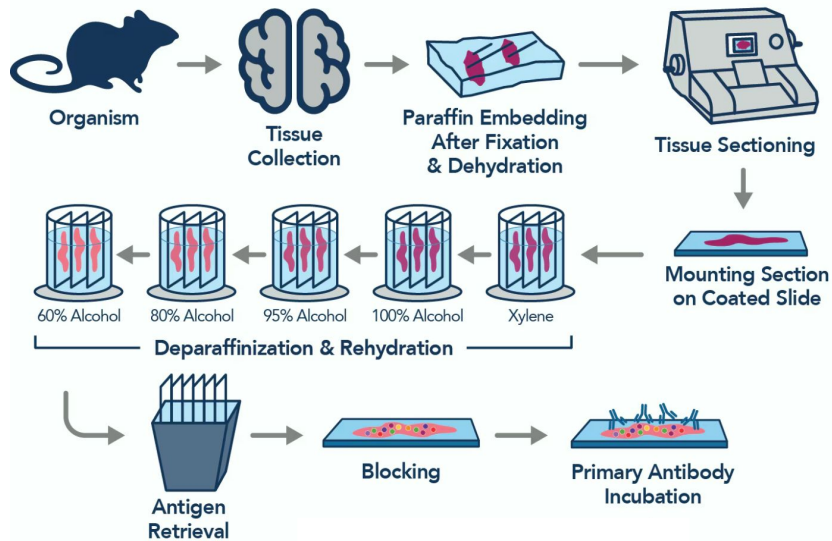
4. Add HRP or AP conjugated secondary antibody into each well and develop colorimetric reaction with appropriate substrate



5. Read absorbance in spectrophotometer with appropriate filter and quantitate relative antigen levels

ELISA

¿Cómo evaluamos?



Inmunofluorescencia

Conectoma

ALLEN BRAIN ATLAS
DATA PORTAL

HOME

MOUSE CONNECTIVITY

TOOLS

PROJECTION BDA/AAV COMPARISON TRANSGENIC CHARACTERIZATION REFERENCE DATA BRAIN EXPLORER DOCUMENTATION HELP

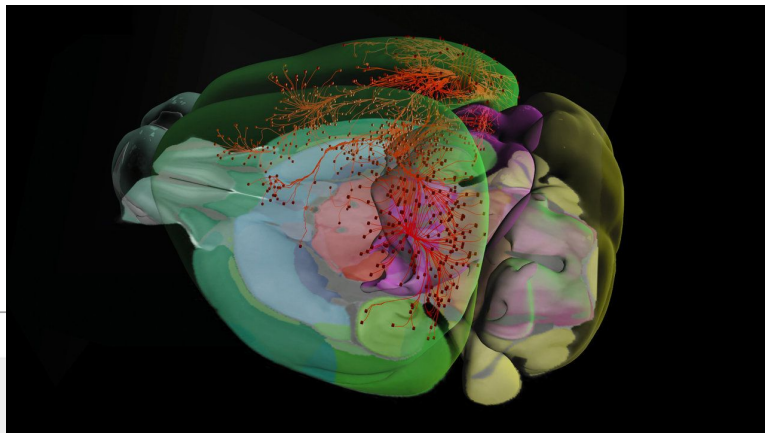
Filter Source Structure(s) ?

- ☒ Source Search
☐ Target Search
☐ Spatial Search

Filter Mouse Line

Filter Tracer Type

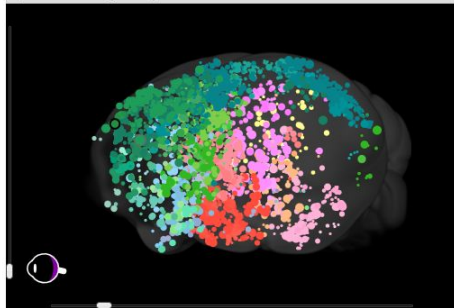
Primary Structure Only ☒ Intrinsic Signal Images ☐



Search Results

Permalink

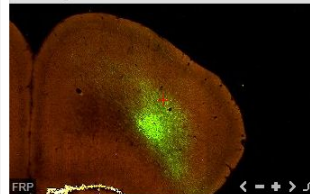
Injection Sites - Showing 2917 Experiments



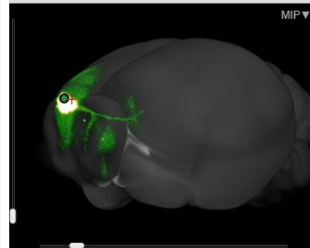
Injection Structure(s)	Mouse Line...	Trace...	Inj Site Vol...
☒ FRP - MOs, ORBl	Cux2-IRES-Cre	EGFP	0.094
☐ FRP - MOs, ORBl	Htr2a-Cre_K...	EGFP	0.190
☐ FRP - MOs	C57BL/6J	EGFP	0.156
☐ FRP - MOs, ORBl	Syt6-Cre_K11...	EGFP	0.017
☐ MOp - MOs	C57BL/6J	EGFP	0.371
☐ MOp - MOs, SSP-II, SSP-tr, RSPagl	C57BL/6J	EGFP	1.015
☐ MOp - SSP-II	C57BL/6J	EGFP	0.173
☐ MOp - MOs	Efr3a-Cre_NO...	EGFP	0.316
☐ MOp	C57BL/6J	EGFP	0.344

Experiment 286299886 - FRP

Section Images



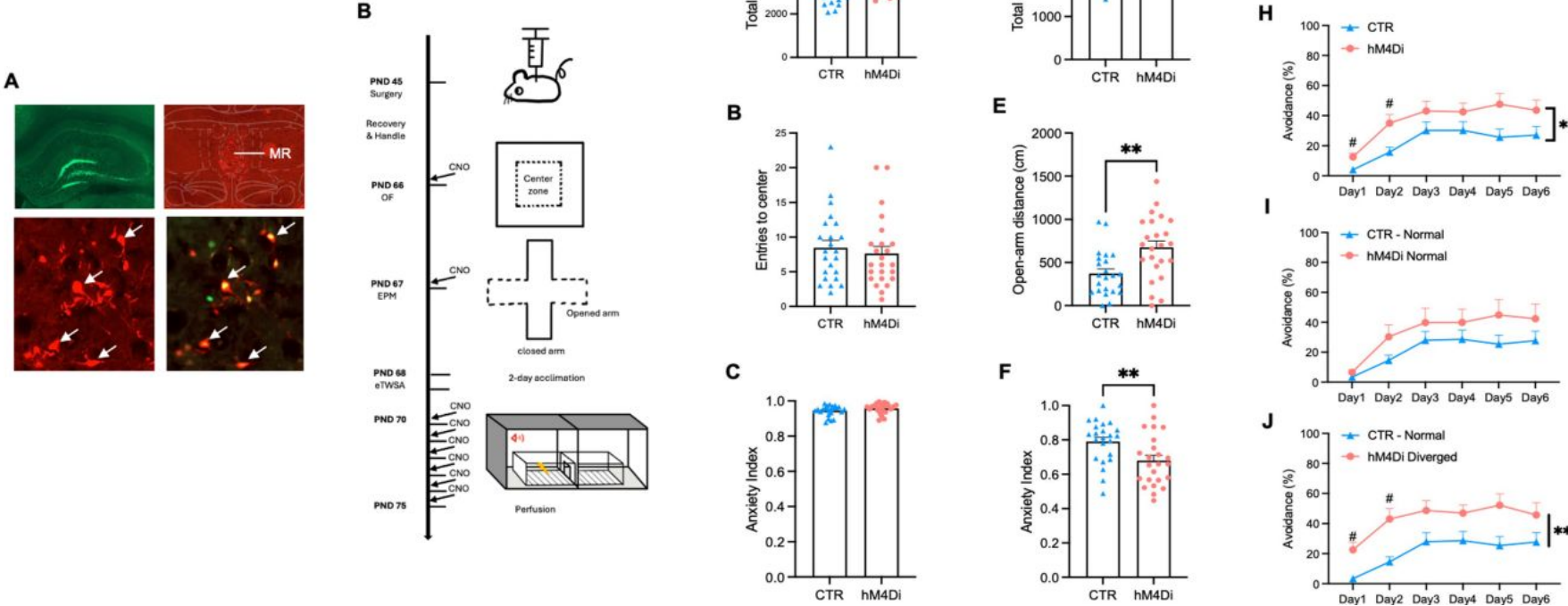
Projection Density

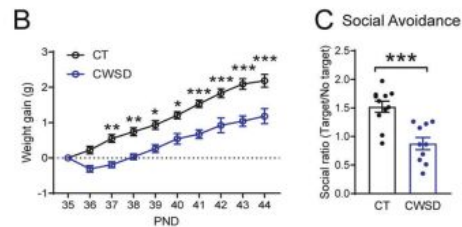
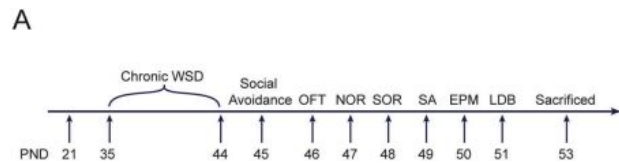


Transgenic Line

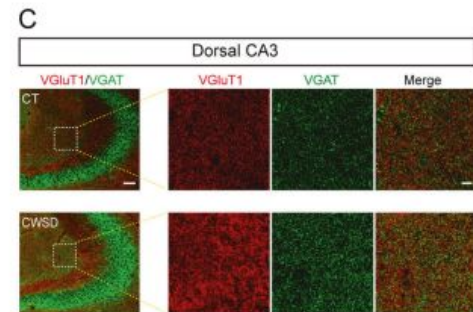
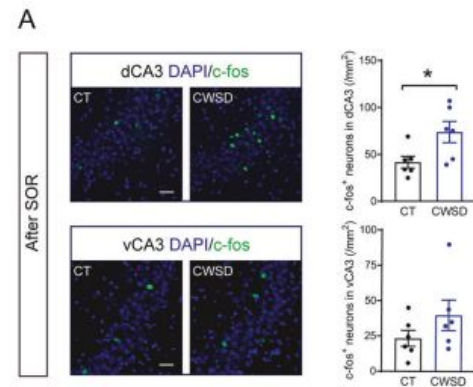
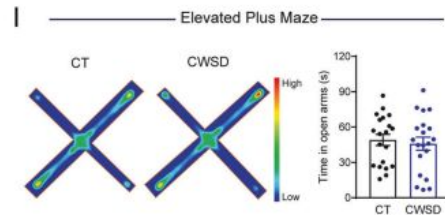
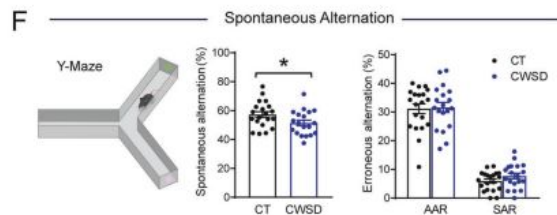
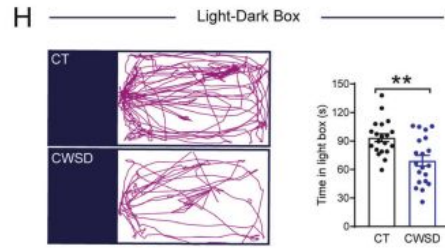
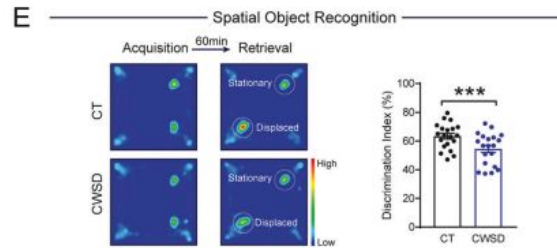
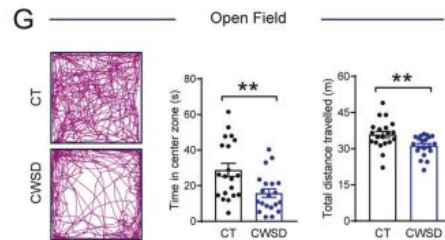
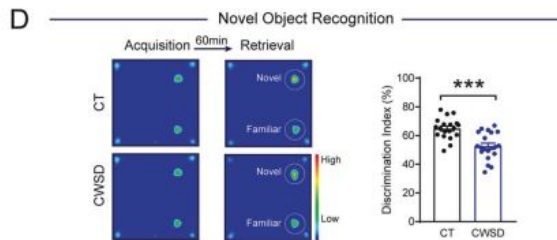
Cux2-IRES-Cre

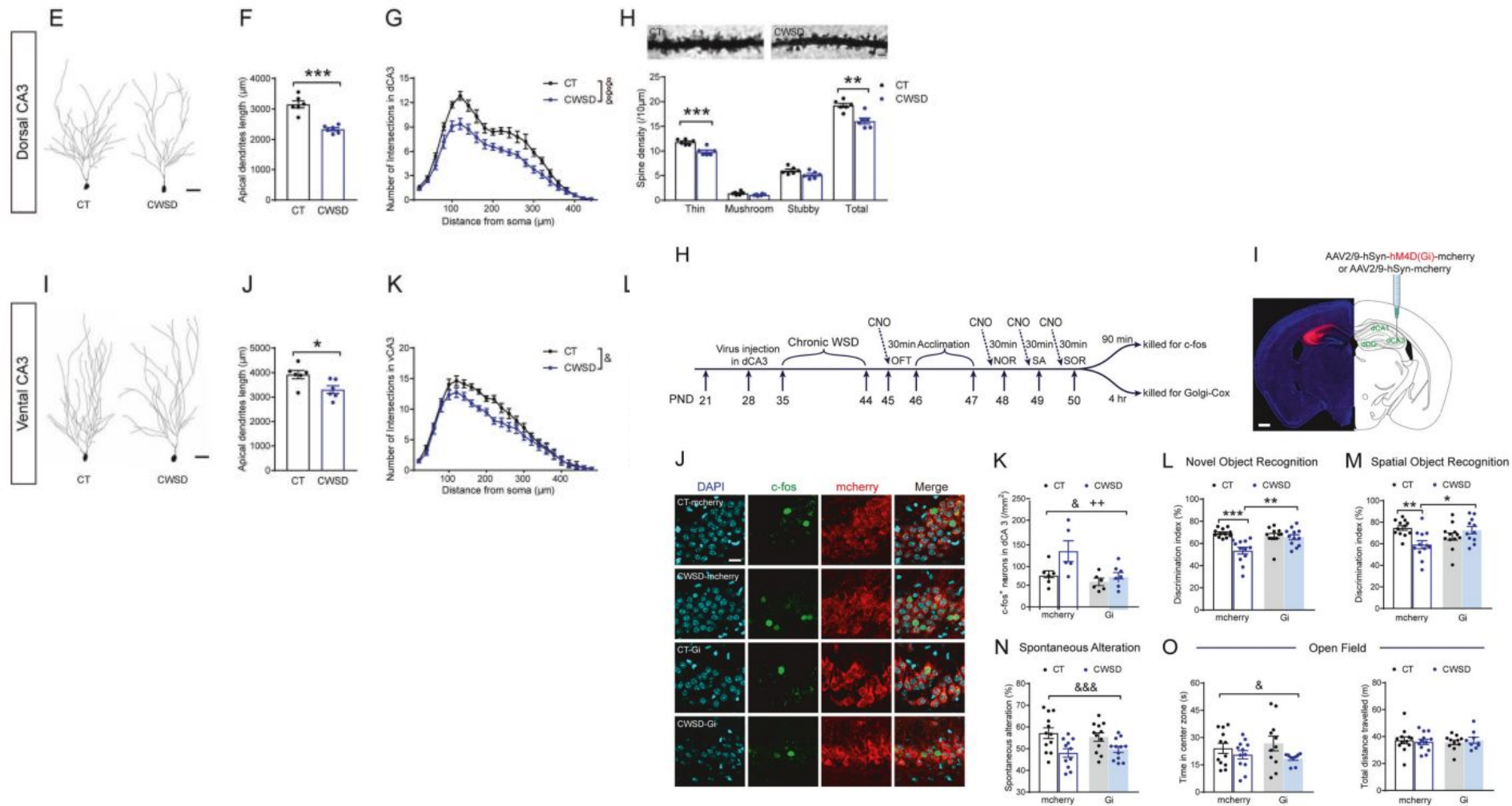
Inhibition of MR-dorsal DG projections induced anxiolytic-like effect and improved learning ability under stress



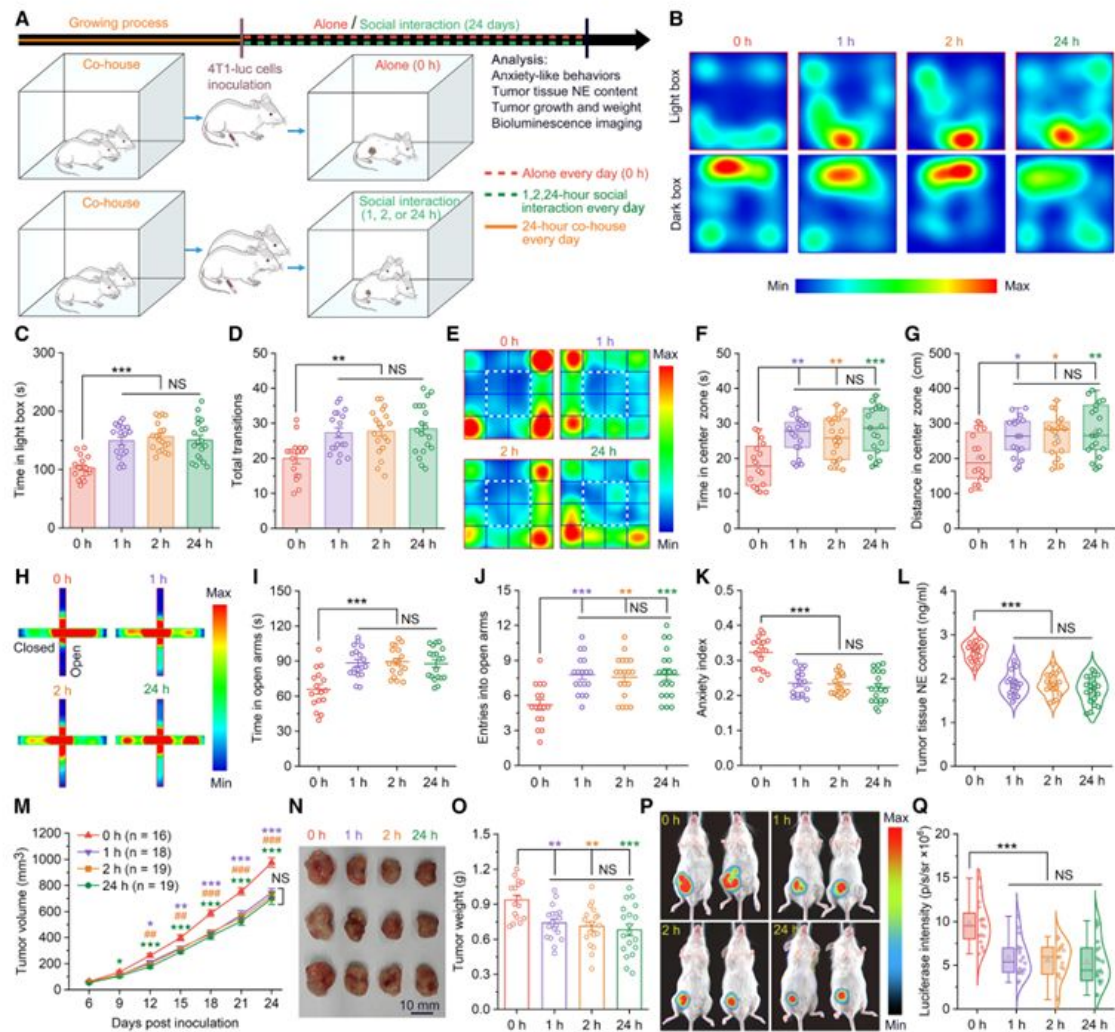


Dorsal CA3 overactivation mediates witnessing stress-induced recognition memory deficits in adolescent male mice

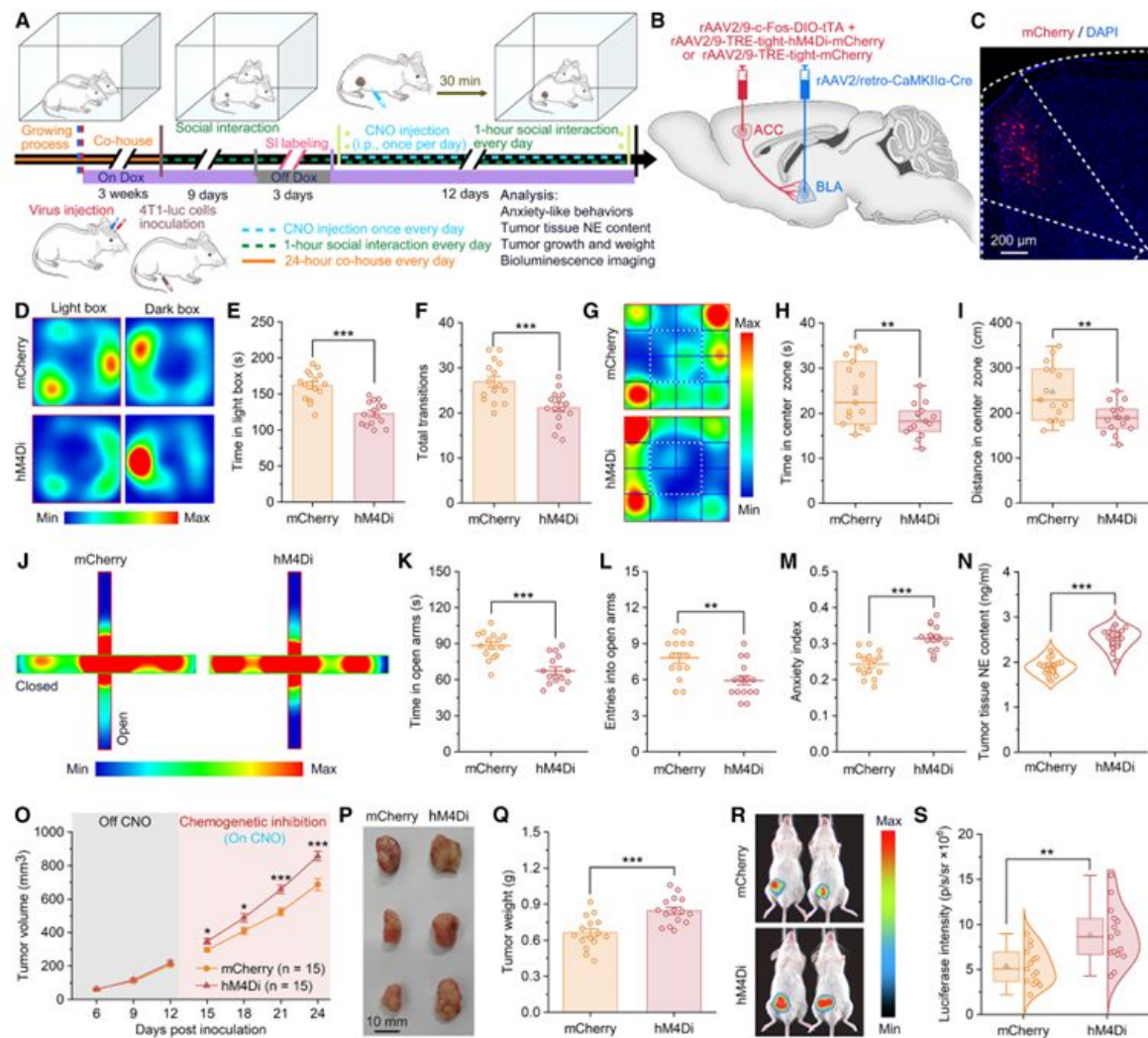


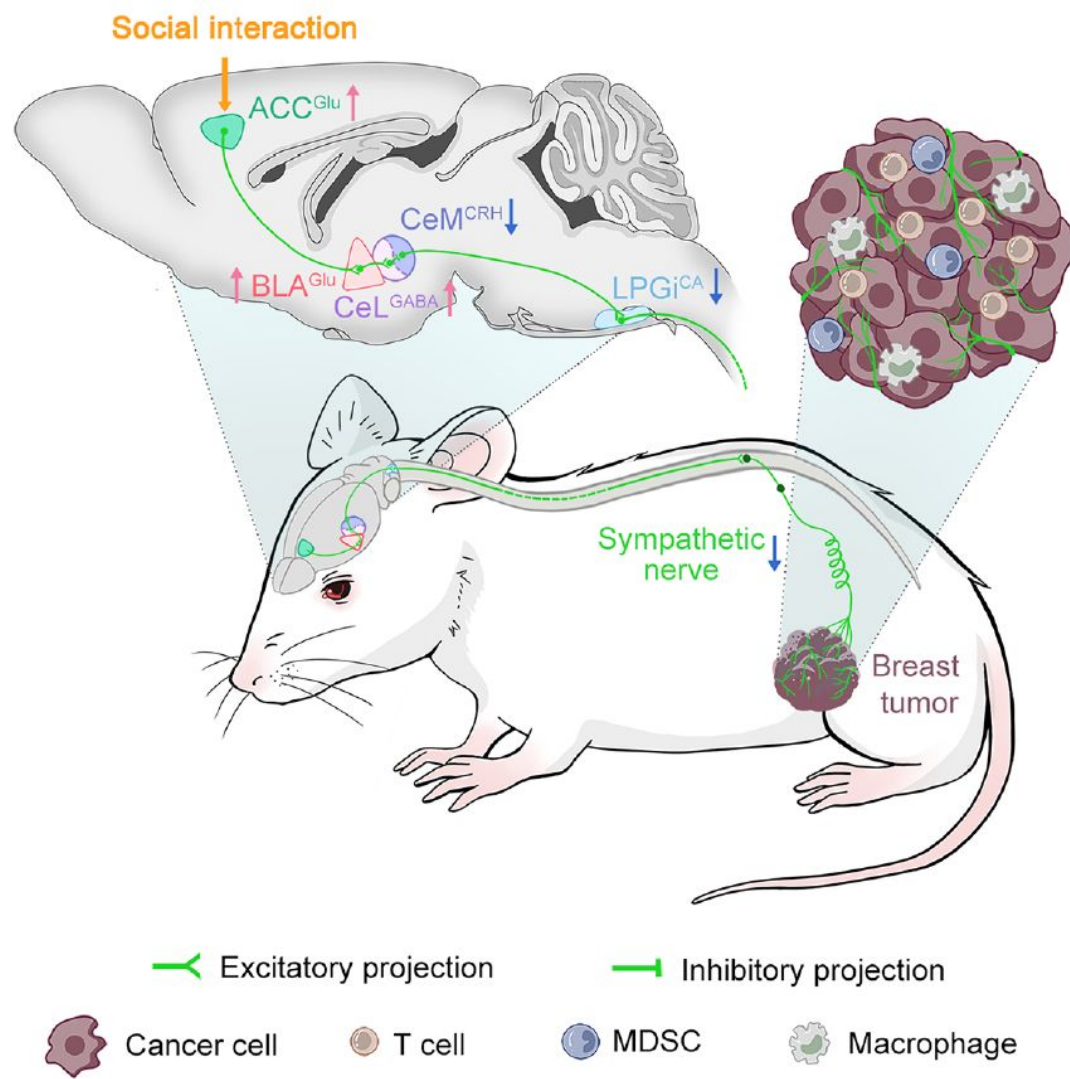


Social interaction in mice suppresses breast cancer progression via a corticoamygdala neural circuit



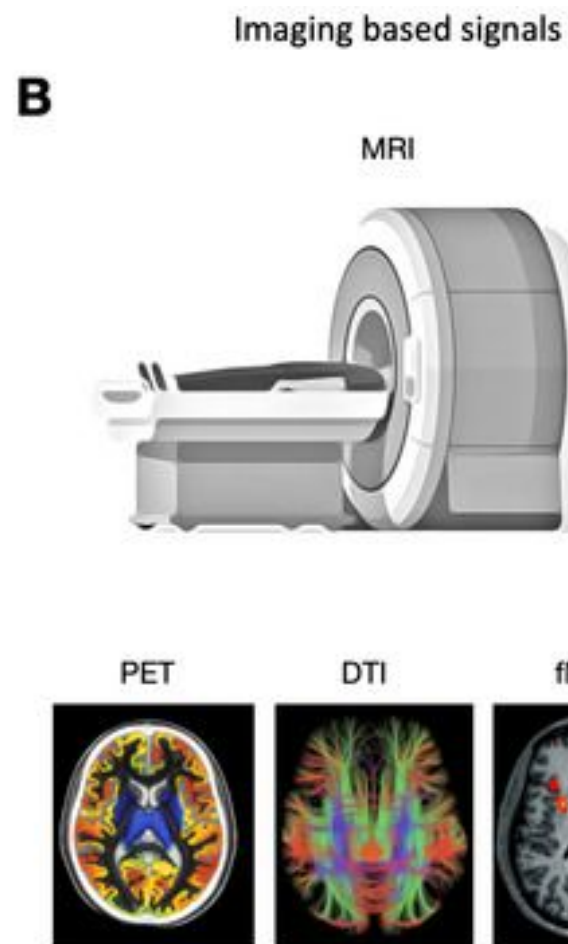
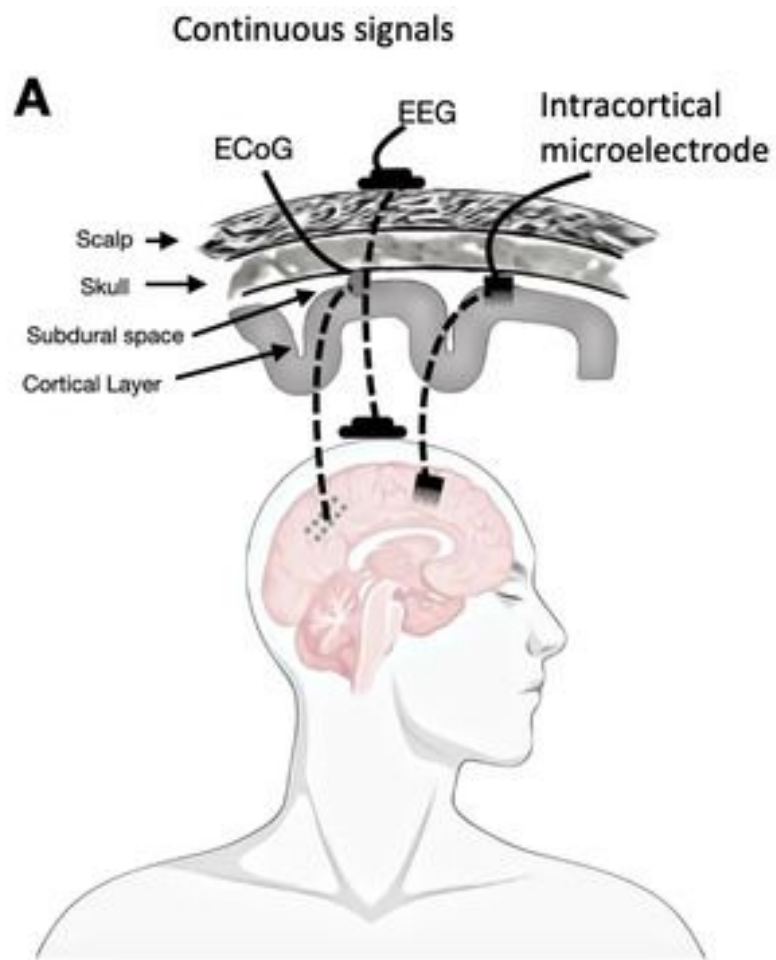
Social interaction in mice suppresses breast cancer progression via a corticoamygdala neural circuit



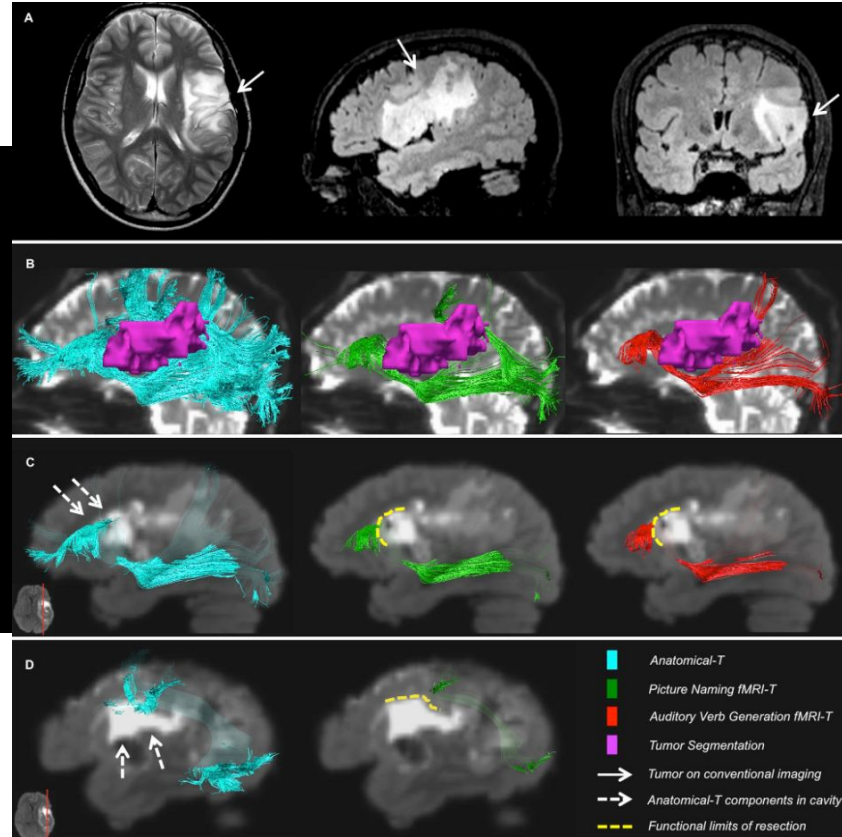
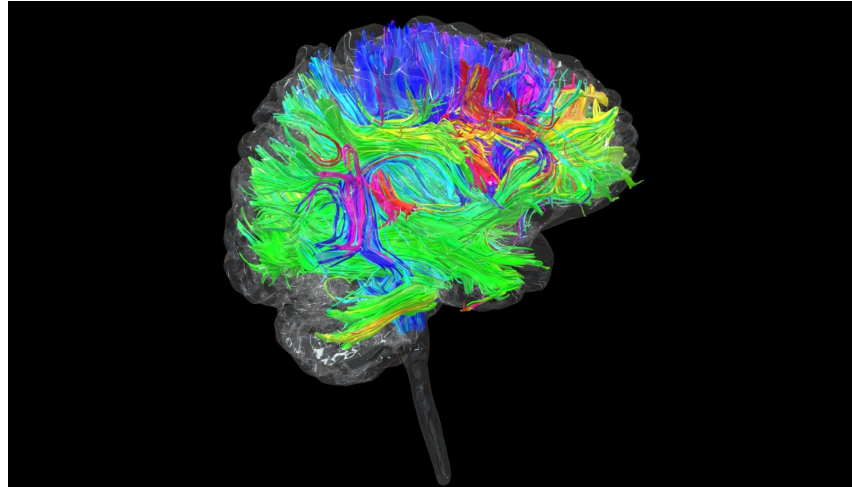


¿Y los animales humanos?



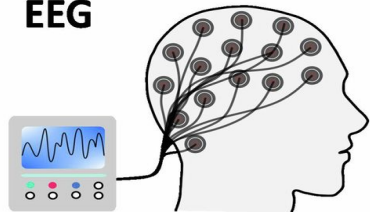


Neuroimágenes



Registros de actividad cerebral

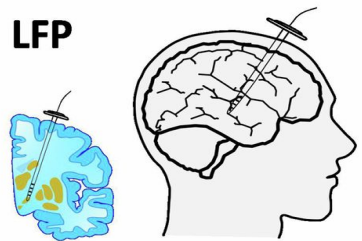
EEG



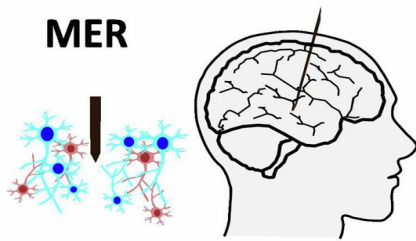
MEG



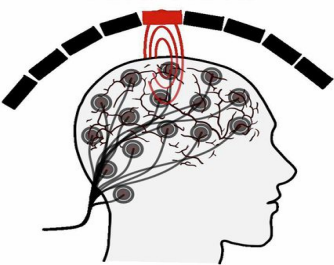
LFP



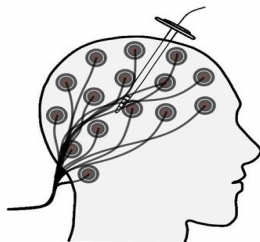
MER



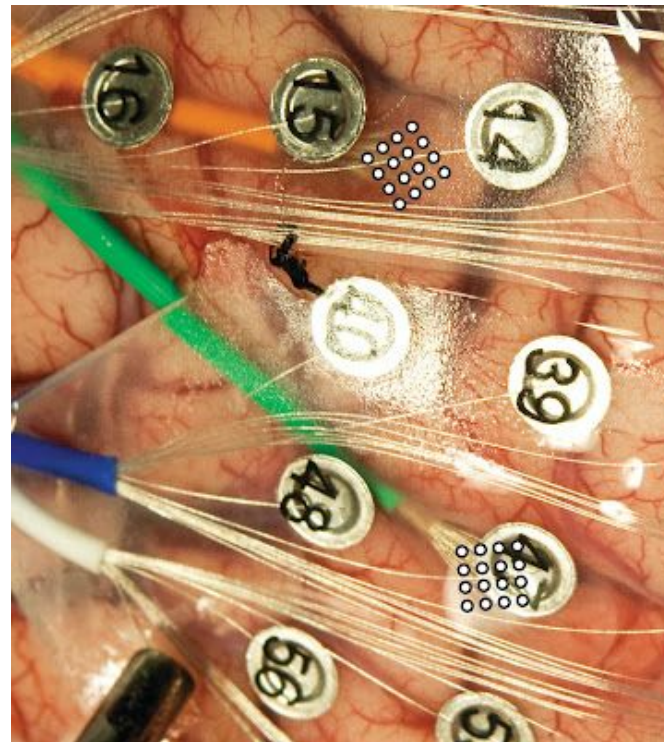
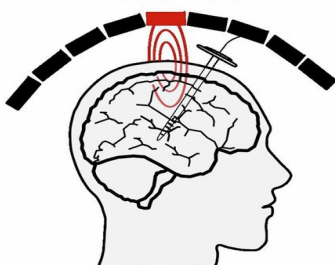
EEG-MEG



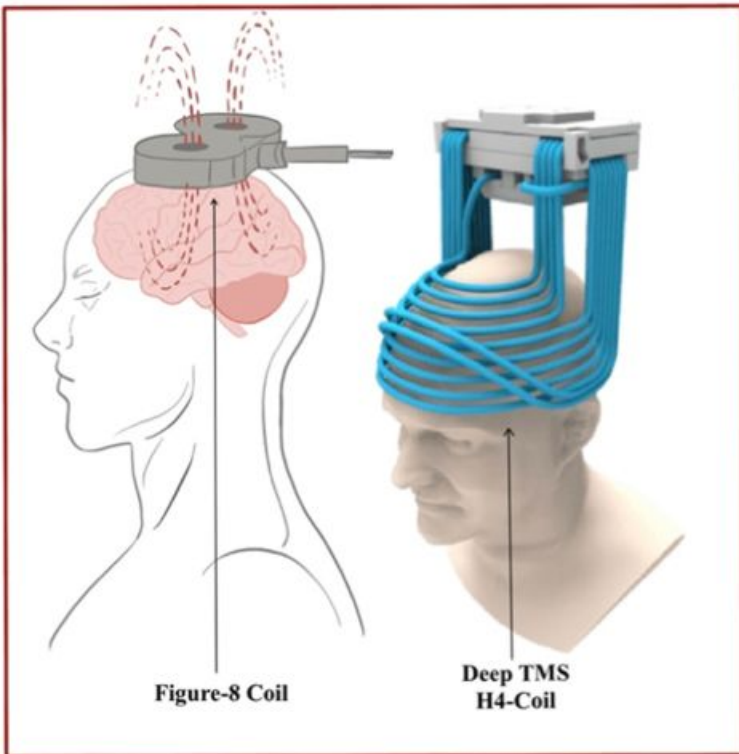
EEG-LFP



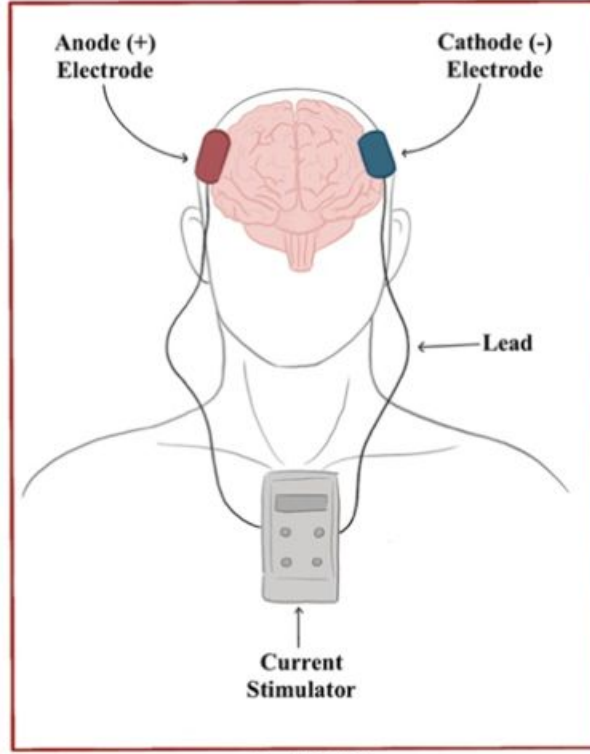
MEG-LFP



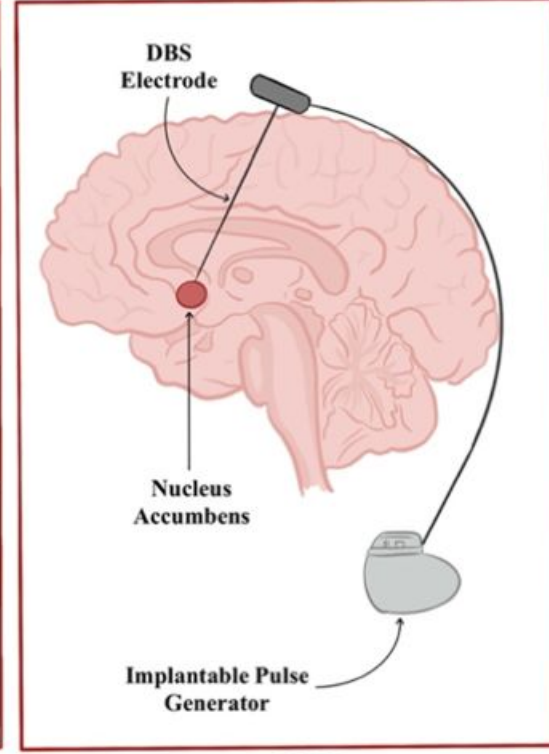
Estimulación cerebral



a. rTMS



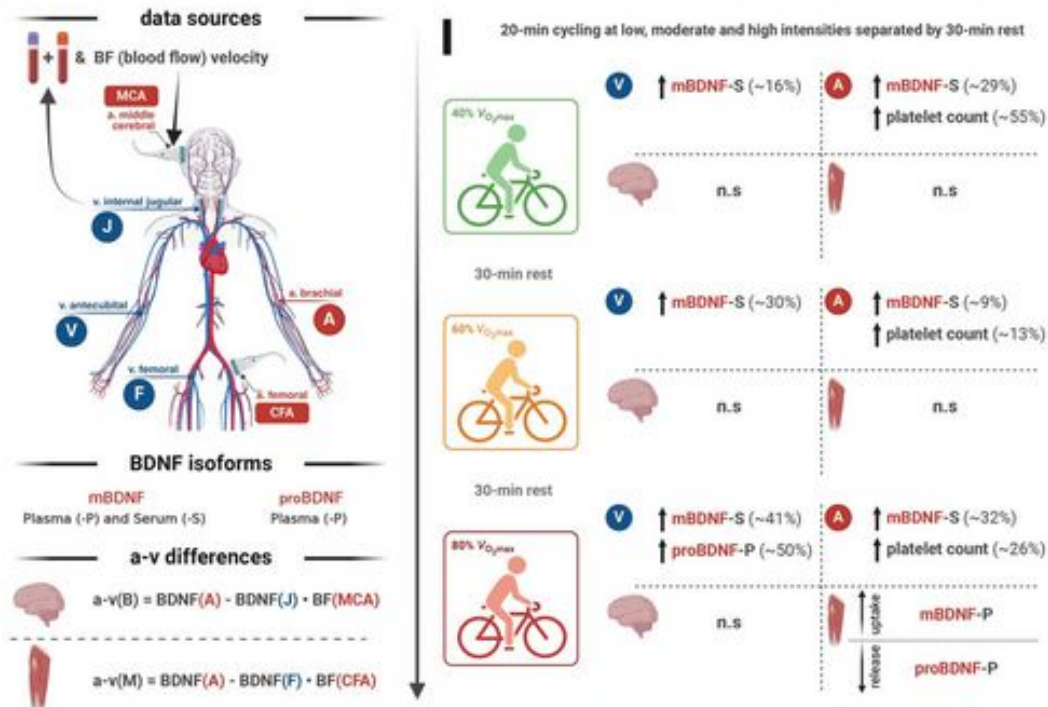
b. tDCS



c. DBS

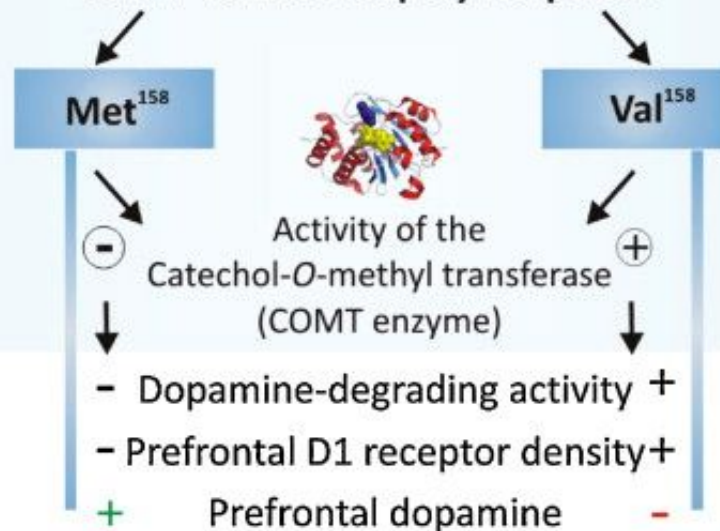
Muestras biológicas in vivo

Arterial-venous differences of BDNF isoforms across the brain and muscle after exercise



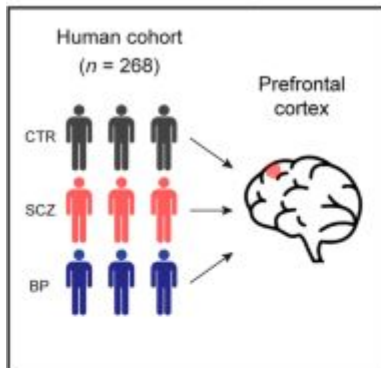
The Journal of
Physiology

COMT Val158Met polymorphism

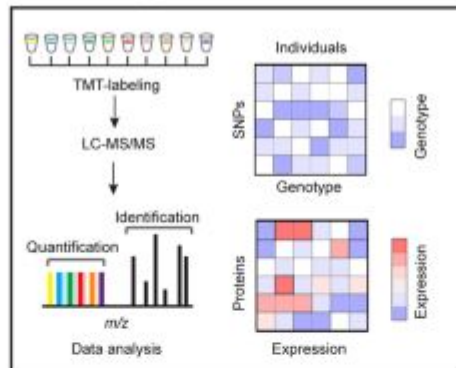


Multiómicas post-mortem

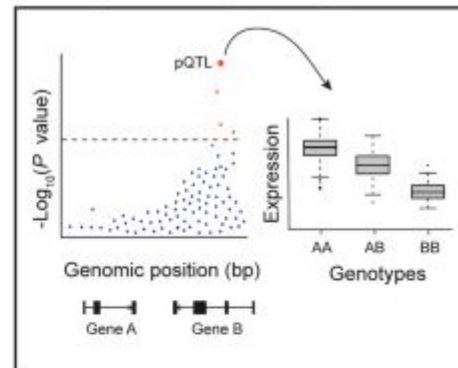
A Human brain samples



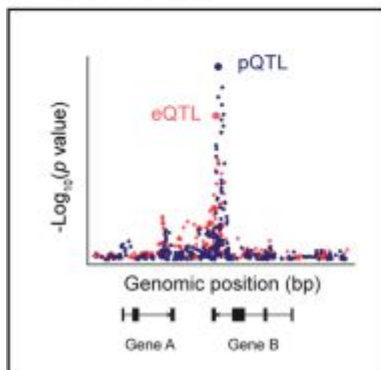
B Protein quantification



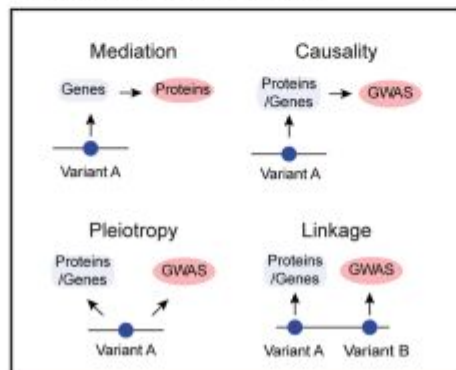
C Regulation of protein expression



D Co-localization



E Mediation and causality



F Prioritization

